

Comparison Between RS67 Vs RS70

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April 24, 2026

Slide 02: Input Files (Roadset & Target Order)

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Remark: Currently we don't process RS5a, RS5b files.

Slide 03: Physics Constants & Normalizations

Parameter	Value
PoT LH ₂	2.7639×10^{17}
PoT LD ₂	1.3197×10^{17}
PoT Flask	5.5905×10^{16}
LH ₂ Target Density	3.5966 mol/cm ²
LD ₂ Target Density	8.0431 mol/cm ²
LH ₂ / LD ₂ Target Length	50.8 cm
Avogadro Constant	6.022×10^{23}
xF Bin Width	0.05
Nuclear Int. Length (LH ₂ /LD ₂)	52.0 / 54.7 g/cm ²

Slide 04: Event Selection Summary

Dimuon Level $|dx| < 0.25$, $\text{mass} \in [4.2, 8.8]$, $x_F \in [-0.1, 0.95]$, $x_T \in [0.05, 0.58]$, $\chi^2 < 18$.

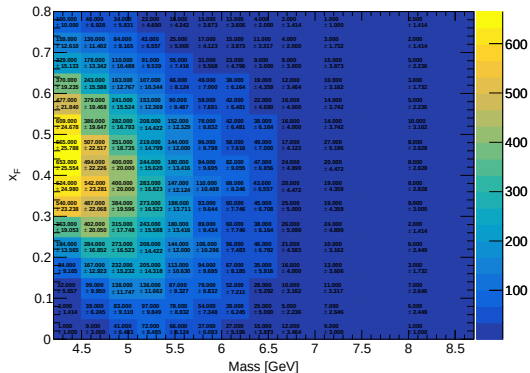
Track Level $\chi^2_{\text{target}} < 15$, $p_{z_{st1}} \in [9, 75]$, Hits > 13 , Station 1/3 y-ratio < 1 .

Occupancy $D1, D2, D3 < 400$ hits each; Total $D1 + D2 + D3 < 1000$.

Physics $Z_{\text{vertex}} \in [-320, -5]$, Track Separation < 270 , $|\cos \theta| < 0.5$.

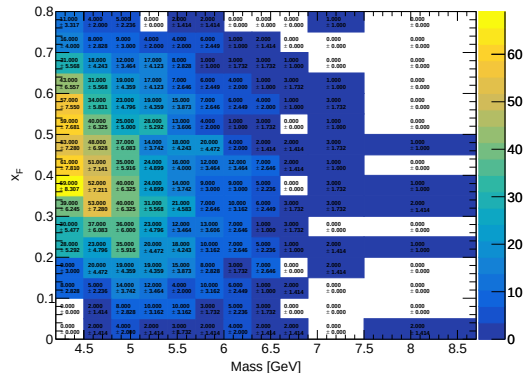
RS67

Total Yield (result) (LH2)



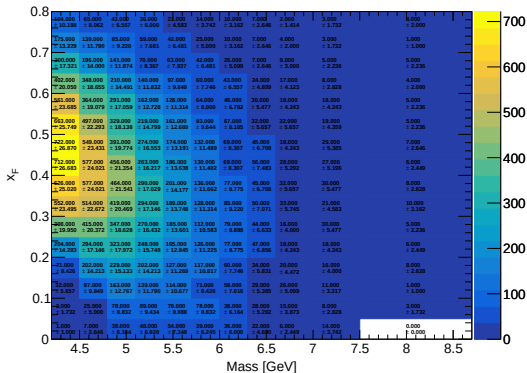
RS70

Total Yield (result) (LH2)



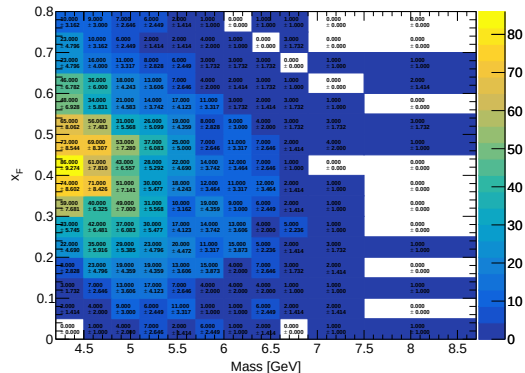
RS67

Total Yield (result) (LD2)



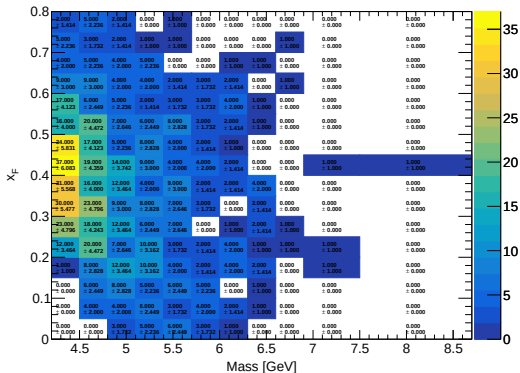
RS70

Total Yield (result) (LD2)



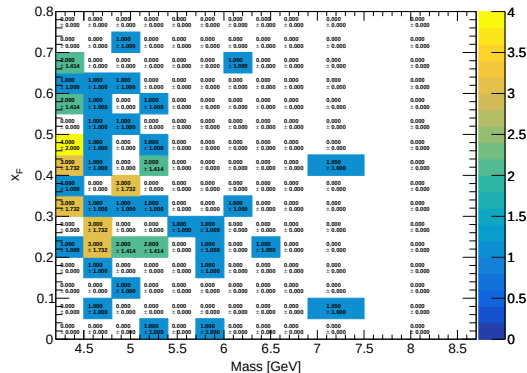
RS67

Total Yield (result) (Flask)



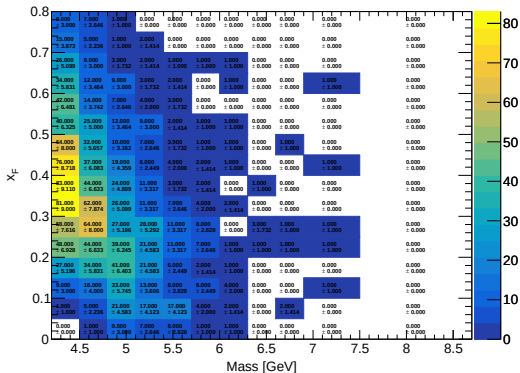
RS70

Total Yield (result) (Flask)



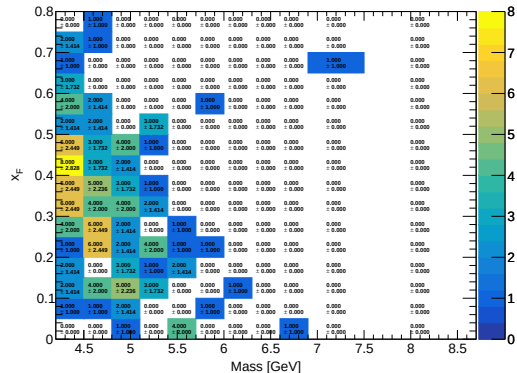
RS67

Mix Yield (result_mix) (LH2)



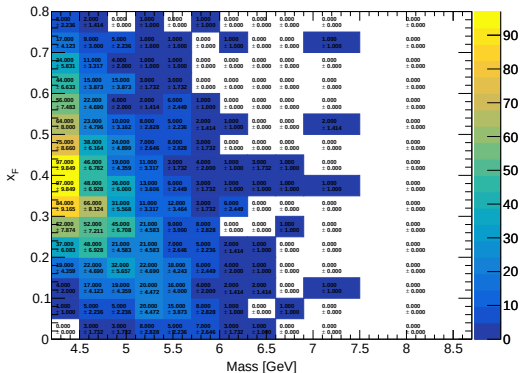
RS70

Mix Yield (result_mix) (LH2)



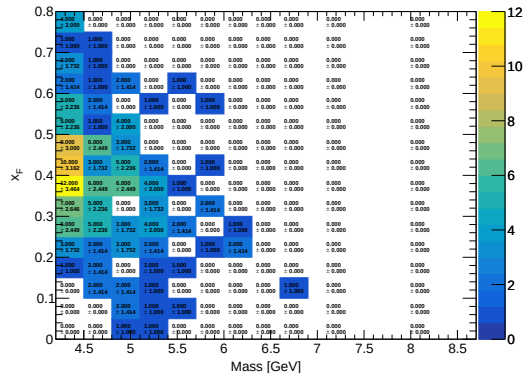
RS67

Mix Yield (result_mix) (LD2)



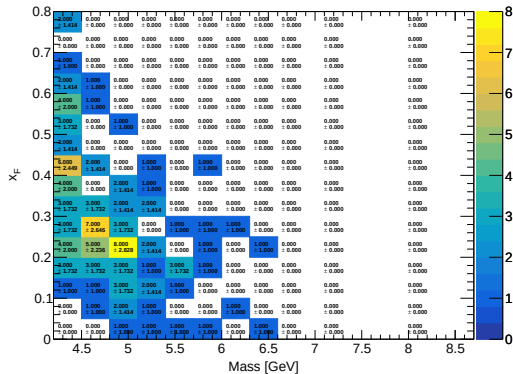
RS70

Mix Yield (result_mix) (LD2)



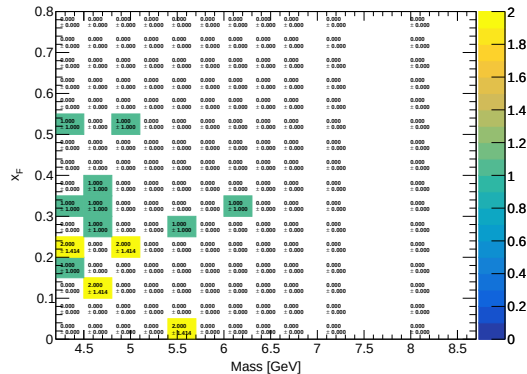
RS67

Mix Yield (result_mix) (Flask)



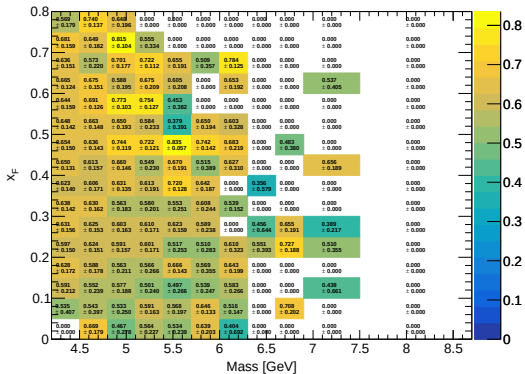
RS70

Mix Yield (result_mix) (Flask)



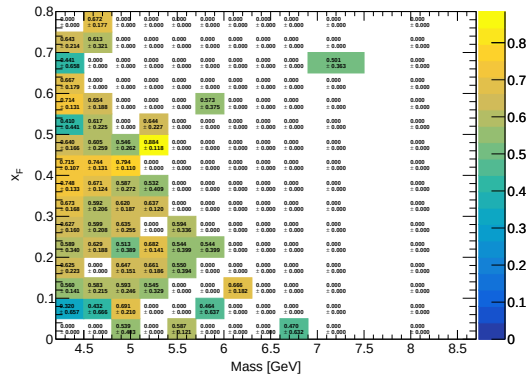
RS67

Avg Reco Eff (Mix) (LH2)



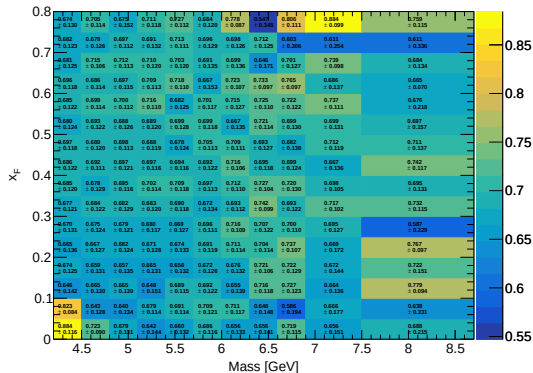
RS70

Avg Reco Eff (Mix) (LH2)



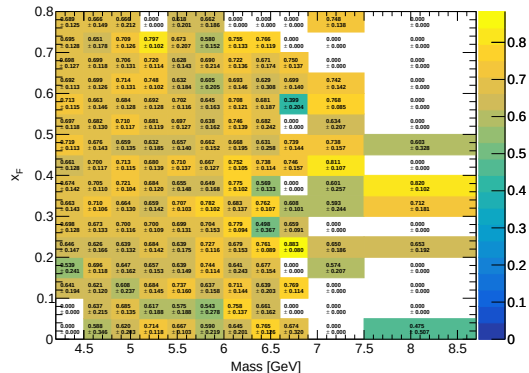
RS67

Avg Reco Eff (Total) (LH2)



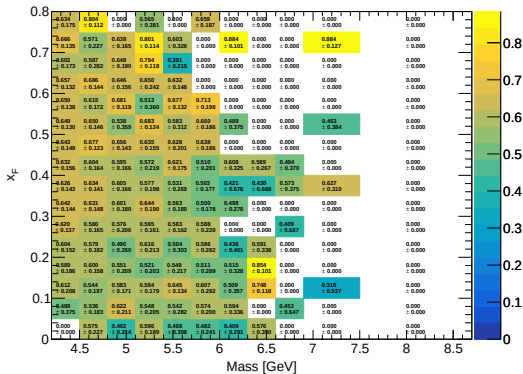
RS70

Avg Reco Eff (Total) (LH2)



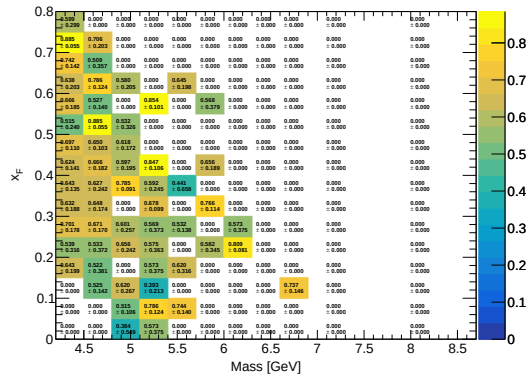
RS67

Avg Reco Eff (Mix) (LD2)



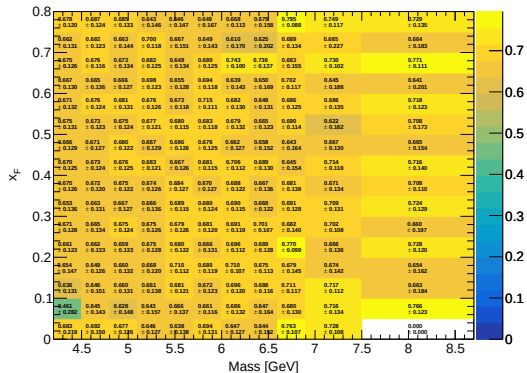
RS70

Avg Reco Eff (Mix) (LD2)



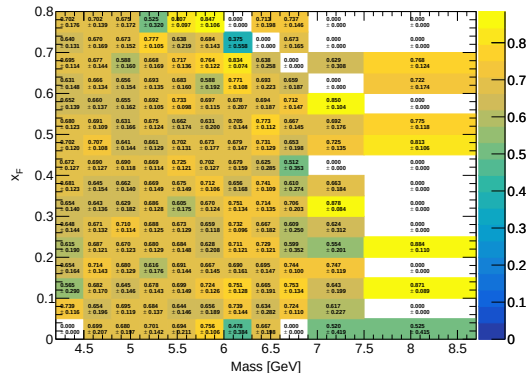
RS67

Avg Reco Eff (Total) (LD2)



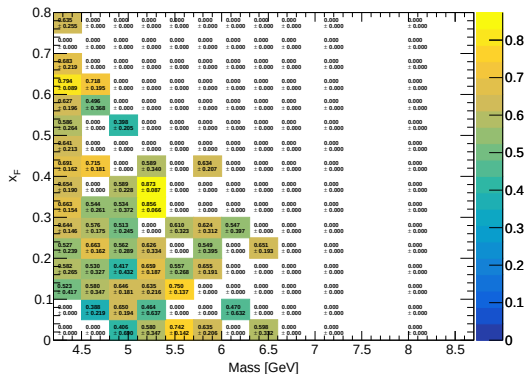
RS70

Avg Reco Eff (Total) (LD2)



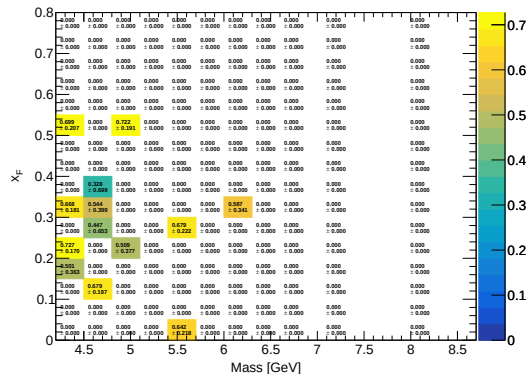
RS67

Avg Reco Eff (Mix) (Flask)



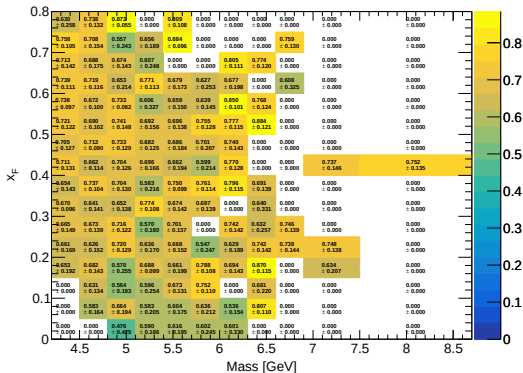
RS70

Avg Reco Eff (Mix) (Flask)



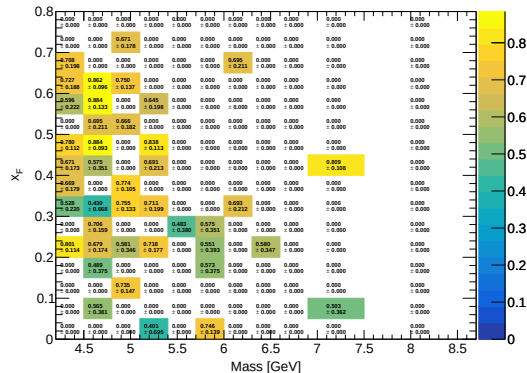
RS67

Avg Reco Eff (Total) (Flask)



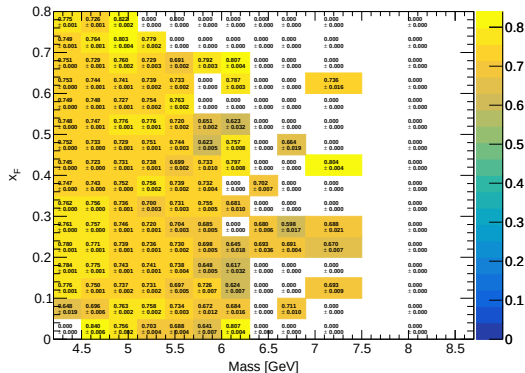
RS70

Avg Reco Eff (Total) (Flask)



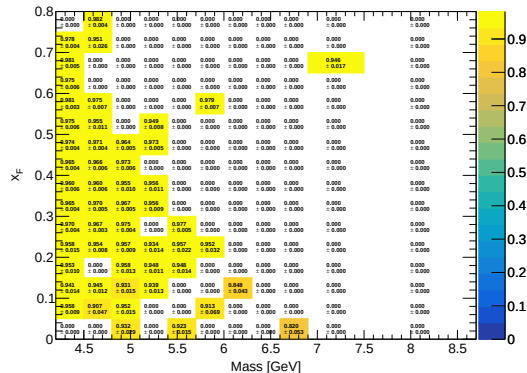
RS67

Avg Hodo Eff (Mix) (LH2)



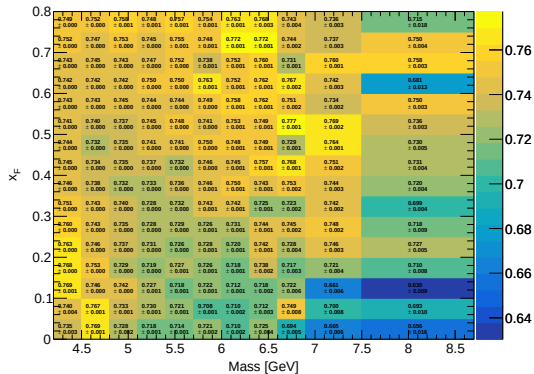
RS70

Avg Hodo Eff (Mix) (LH2)



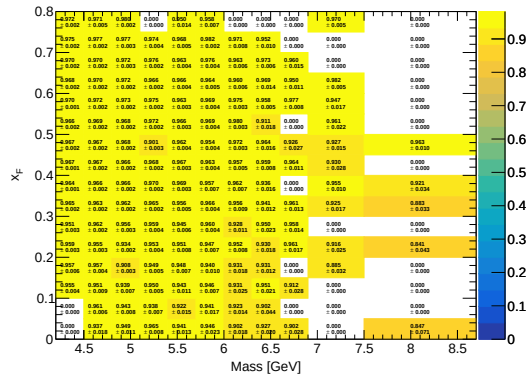
RS67

Avg Hodo Eff (Total) (LH2)



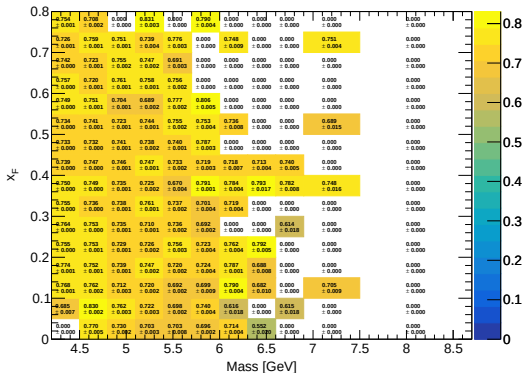
RS70

Avg Hodo Eff (Total) (LH2)



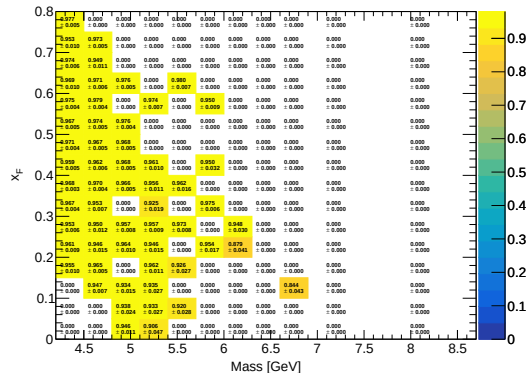
RS67

Avg Hodo Eff (Mix) (LD2)



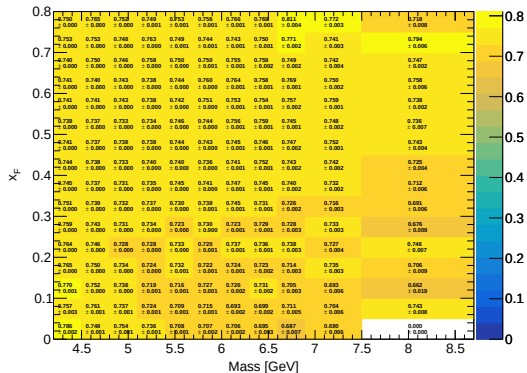
RS70

Avg Hodo Eff (Mix) (LD2)



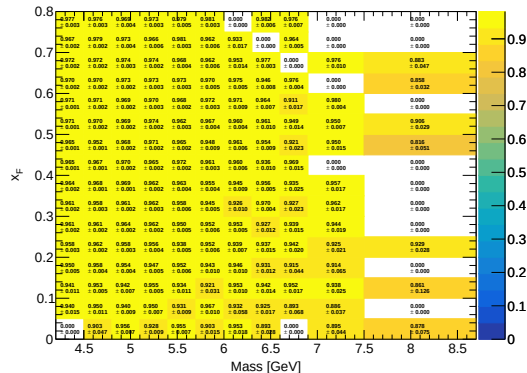
RS67

Avg Hodo Eff (Total) (LD2)



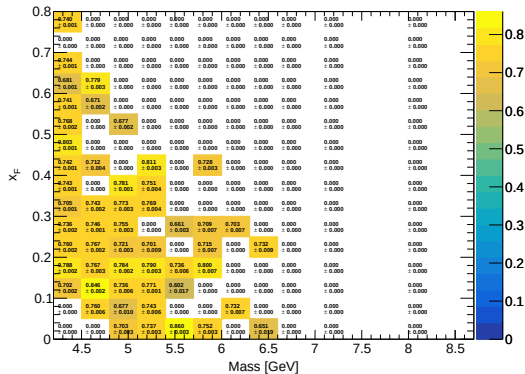
RS70

Avg Hodo Eff (Total) (LD2)



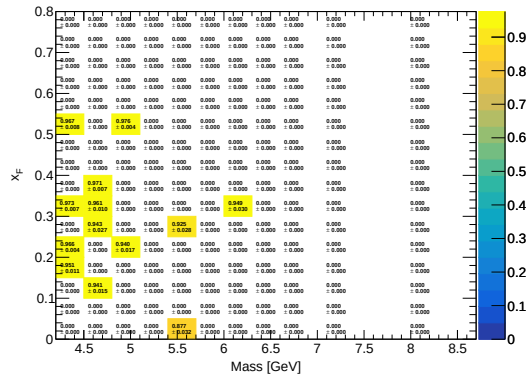
RS67

Avg Hodo Eff (Mix) (Flask)



RS70

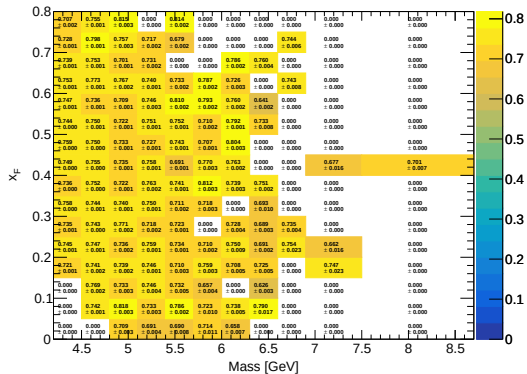
Avg Hodo Eff (Mix) (Flask)



E_total_hodo_Flask

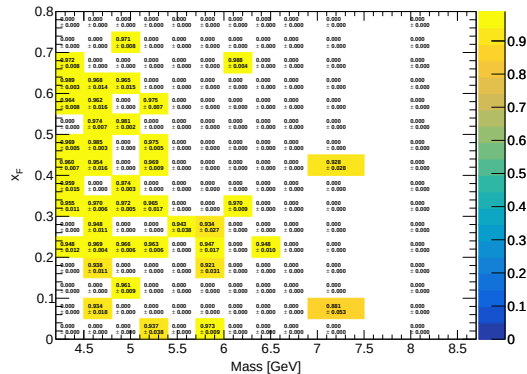
RS67

Avg Hodo Eff (Total) (Flask)



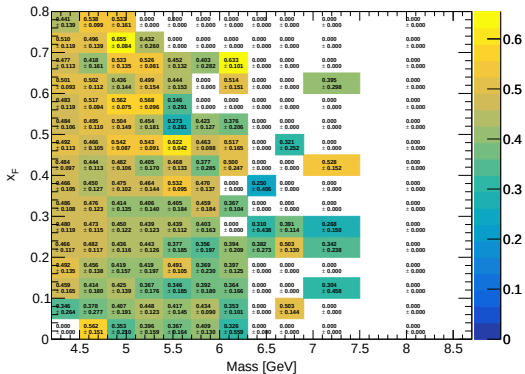
RS70

Avg Hodo Eff (Total) (Flask)



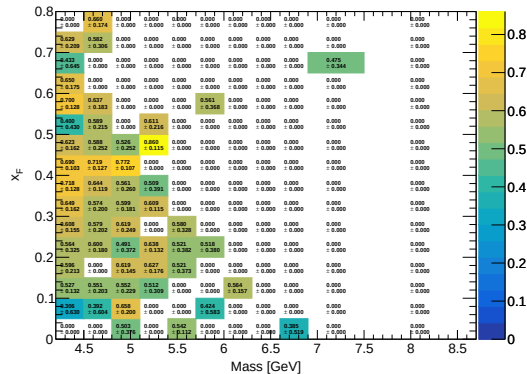
RS67

Avg Final Eff (Mix) (LH2)



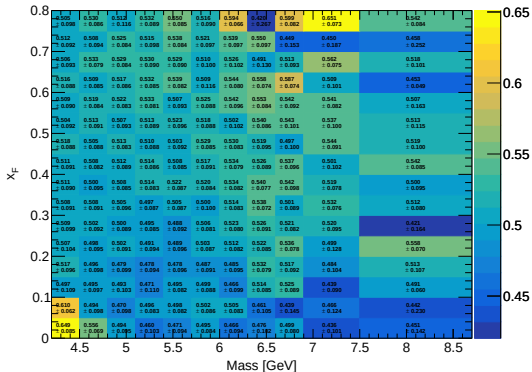
RS70

Avg Final Eff (Mix) (LH2)



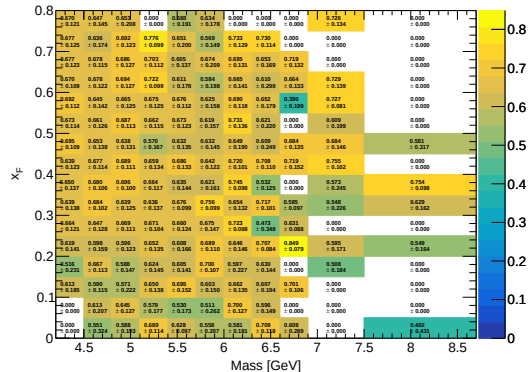
RS67

Avg Final Eff (Total) (LH2)



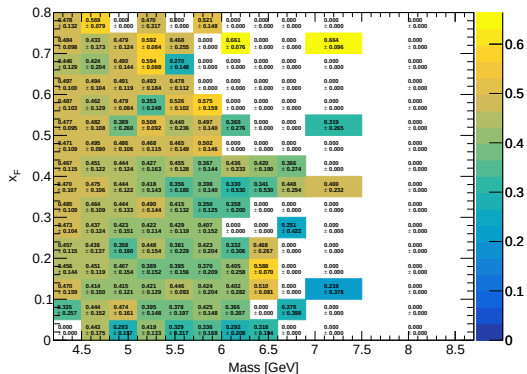
RS70

Avg Final Eff (Total) (LH2)



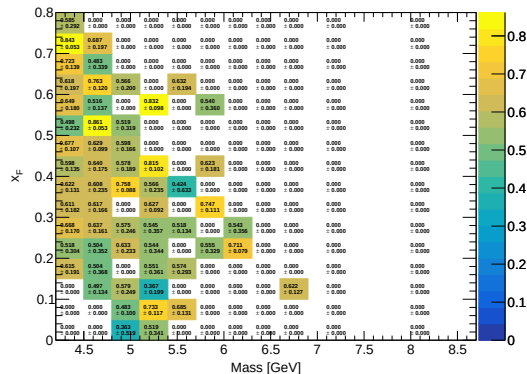
RS67

Avg Final Eff (Mix) (LD2)



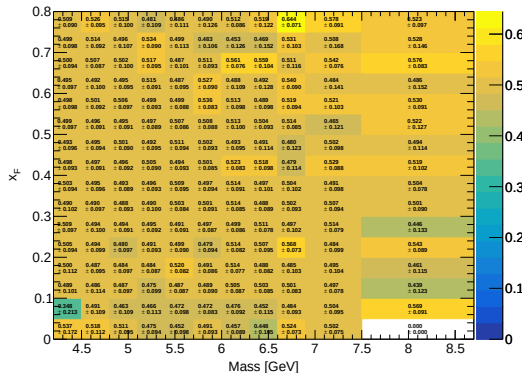
RS70

Avg Final Eff (Mix) (LD2)



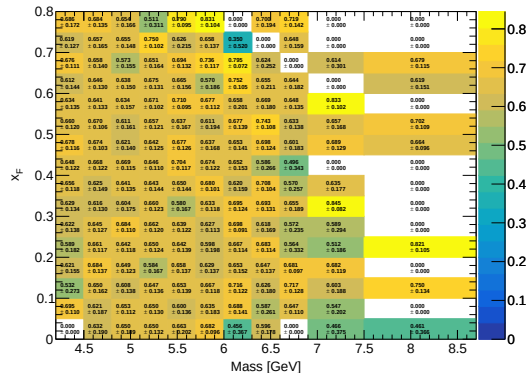
RS67

Avg Final Eff (Total) (LD2)



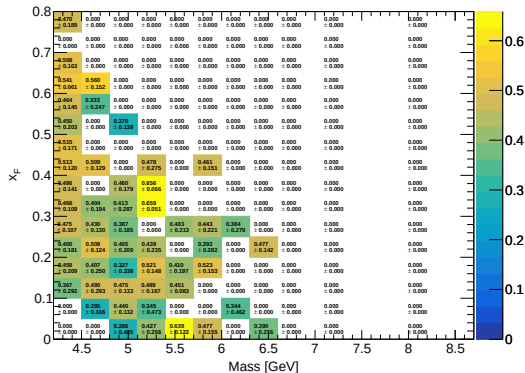
RS70

Avg Final Eff (Total) (LD2)



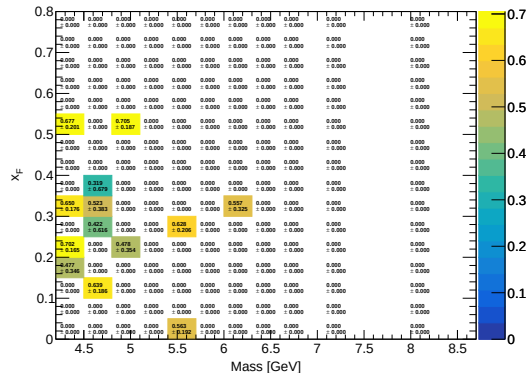
RS67

Avg Final Eff (Mix) (Flask)



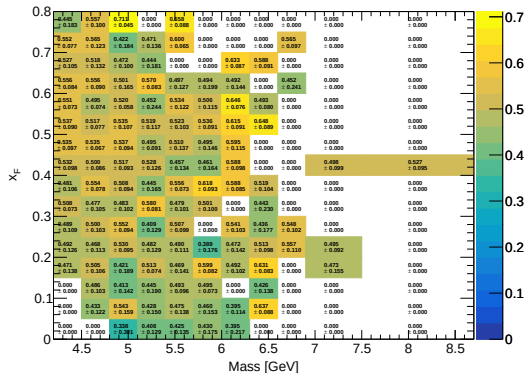
RS70

Avg Final Eff (Mix) (Flask)



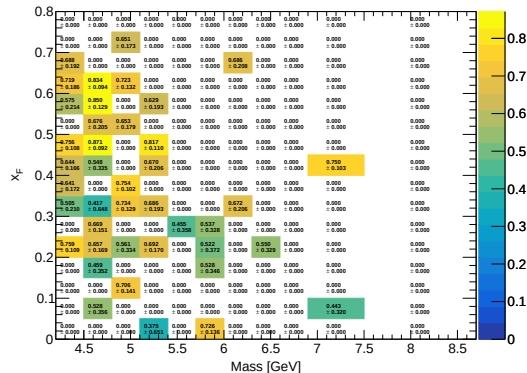
RS67

Avg Final Eff (Total) (Flask)



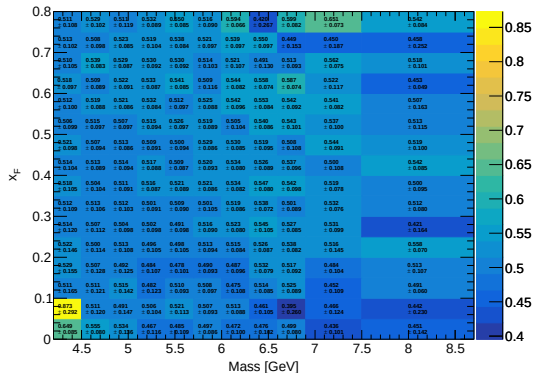
RS70

Avg Final Eff (Total) (Flask)



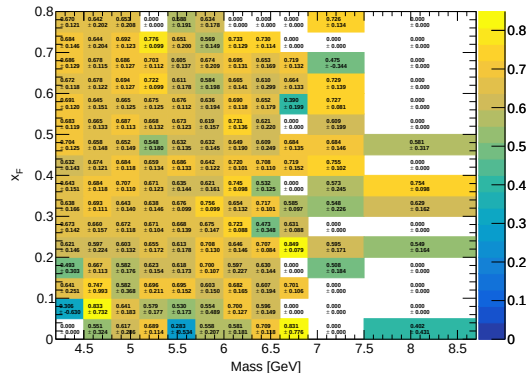
RS67

Avg Signal Efficiency (LH2)



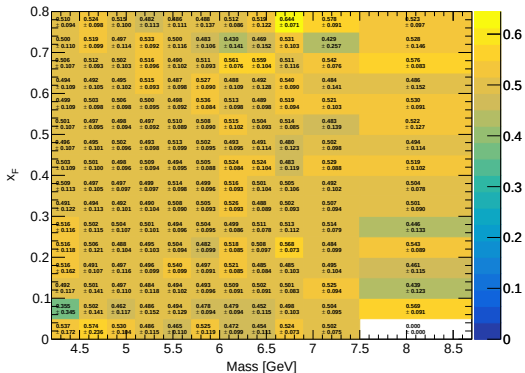
RS70

Avg Signal Efficiency (LH2)



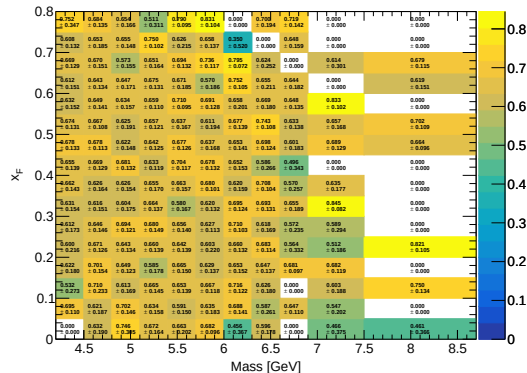
RS67

Avg Signal Efficiency (LD2)



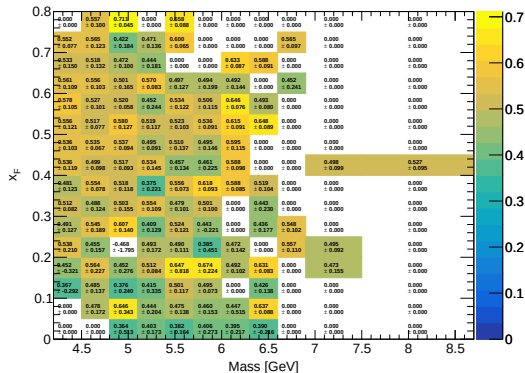
RS70

Avg Signal Efficiency (LD2)



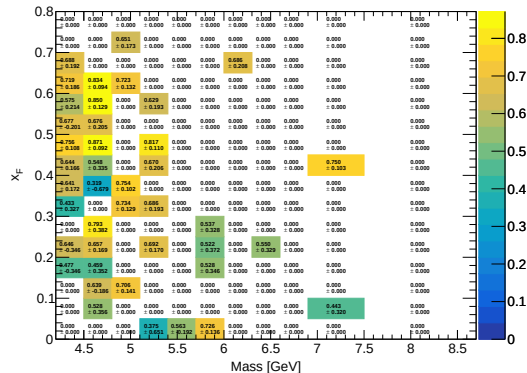
RS67

Avg Signal Efficiency (Flask)



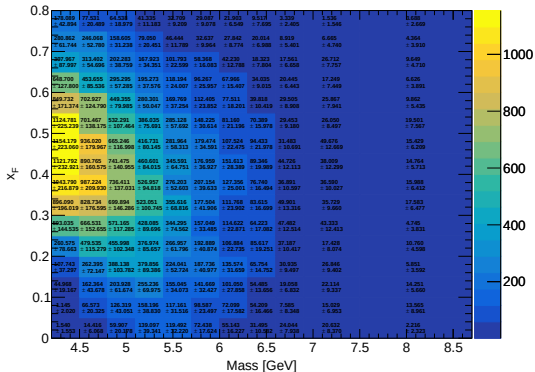
RS70

Avg Signal Efficiency (Flask)



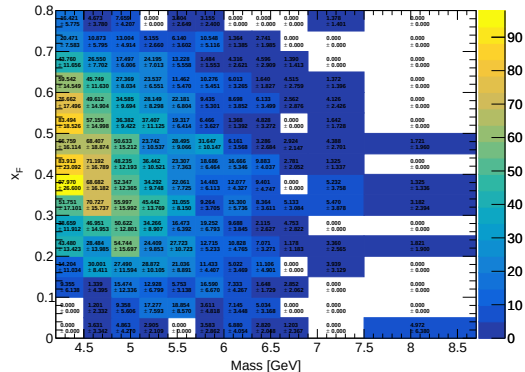
RS67

Corrected Yield (Total Error) (LH2)



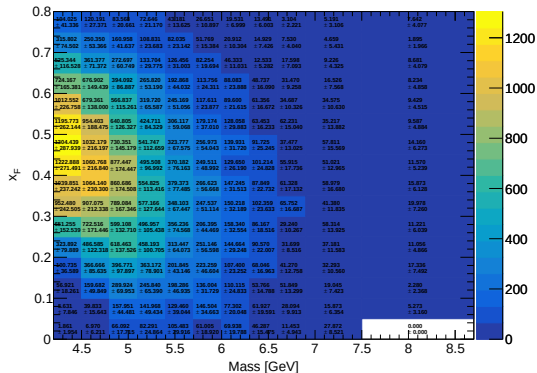
RS70

Corrected Yield (Total Error) (LH2)



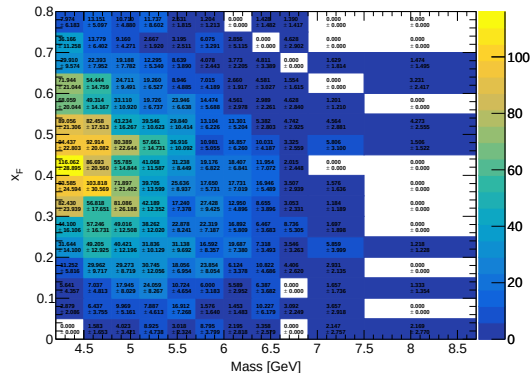
RS67

Corrected Yield (Total Error) (LD2)



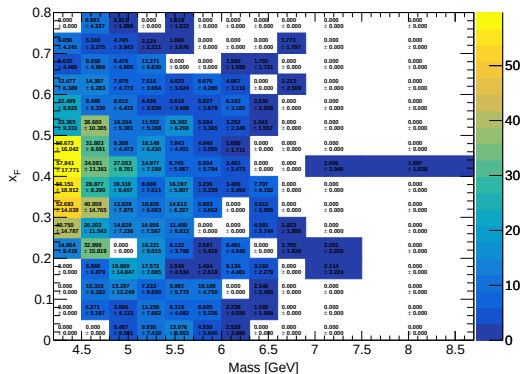
RS70

Corrected Yield (Total Error) (LD2)



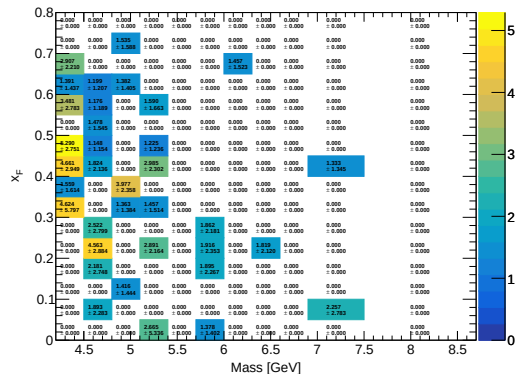
RS67

Corrected Yield (Total Error) (Flask)



RS70

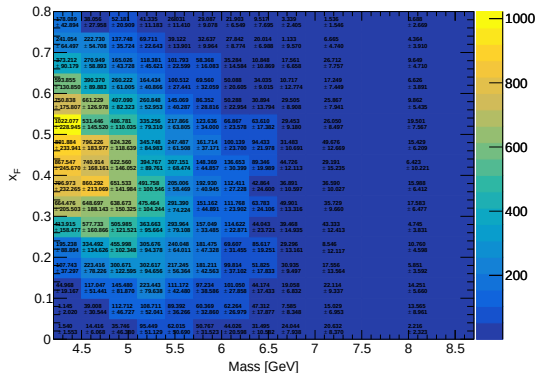
Corrected Yield (Total Error) (Flask)



Y_corrected_Subtracted_LH2

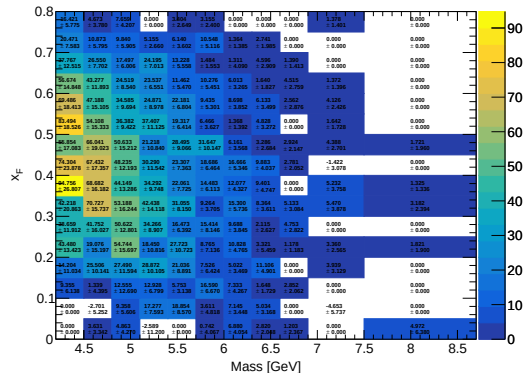
RS67

Corrected Yield (LH2 - Flask)



RS70

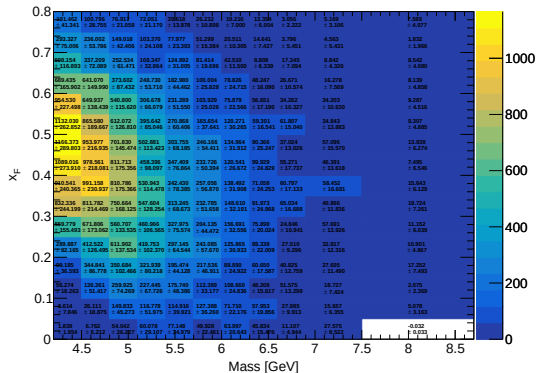
Corrected Yield (LH2 - Flask)



Y_corrected_Subtracted_LD2

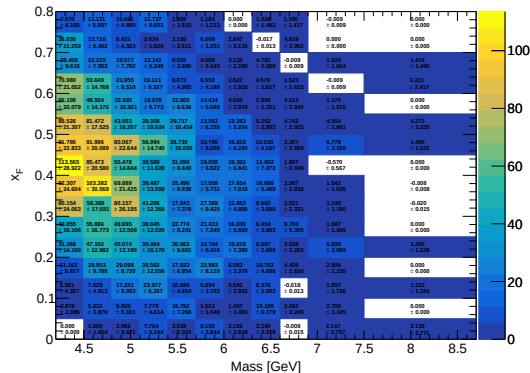
RS67

Corrected Yield (LD2 - LH2 - Flask)



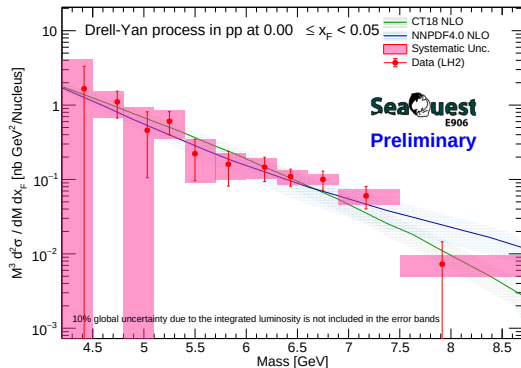
RS70

Corrected Yield (LD2 - LH2 - Flask)

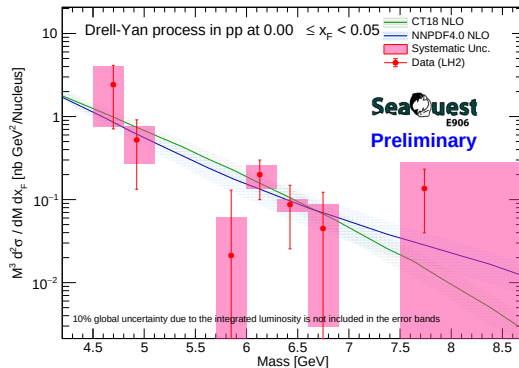


CrossSection LH2 $0.00 \leq x_F < 0.05$ (Centroid)

RS67

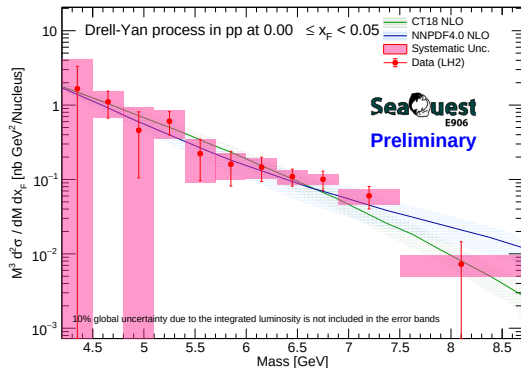


RS70

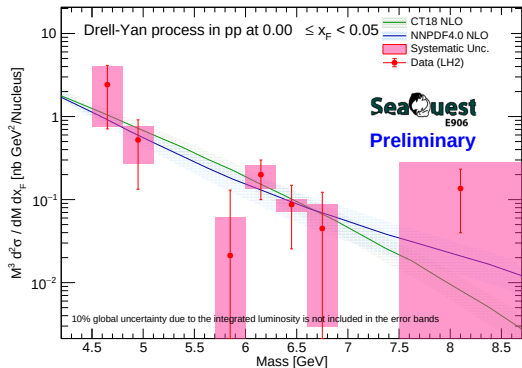


CrossSection LH2 $0.00 \leq x_F < 0.05$ (GeoCenter)

RS67

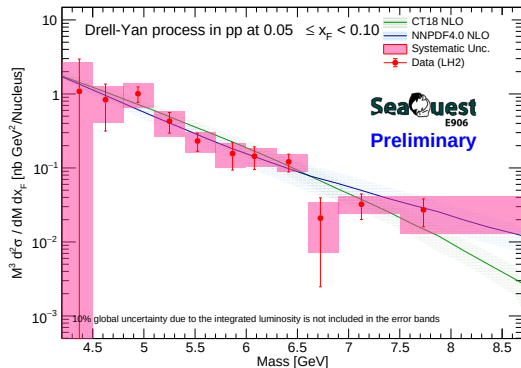


RS70

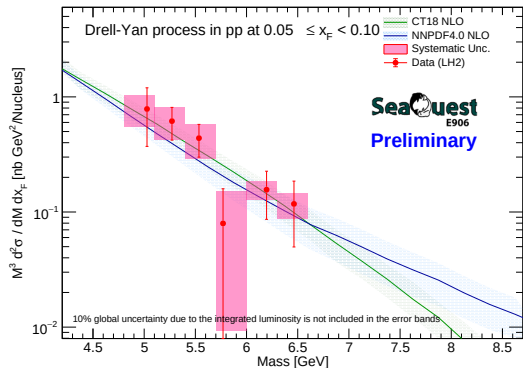


CrossSection LH2 $0.05 \leq x_F < 0.10$ (Centroid)

RS67

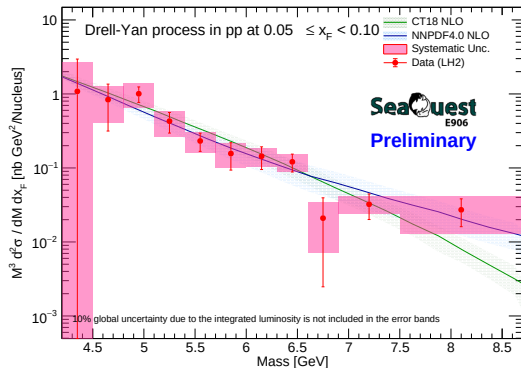


RS70

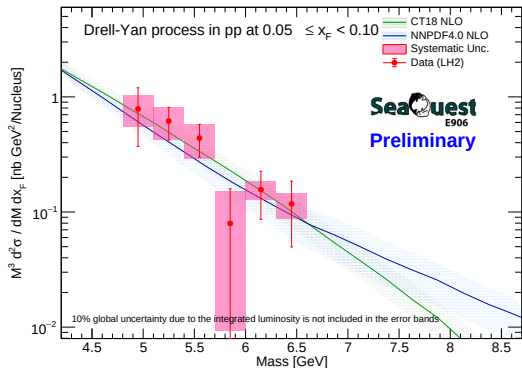


CrossSection LH2 $0.05 \leq x_F < 0.10$ (GeoCenter)

RS67

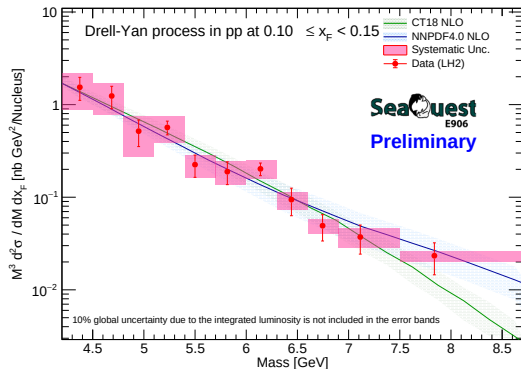


RS70

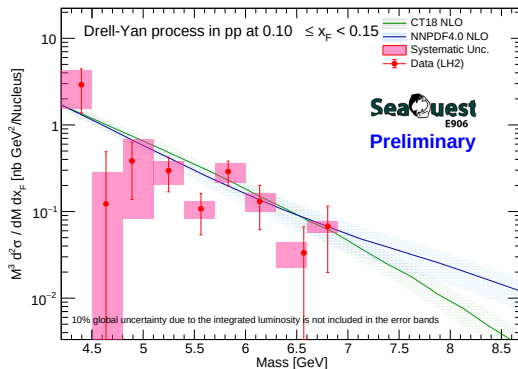


CrossSection LH2 $0.10 \leq x_F < 0.15$ (Centroid)

RS67

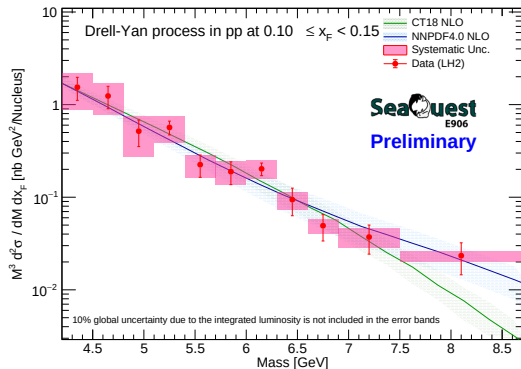


RS70

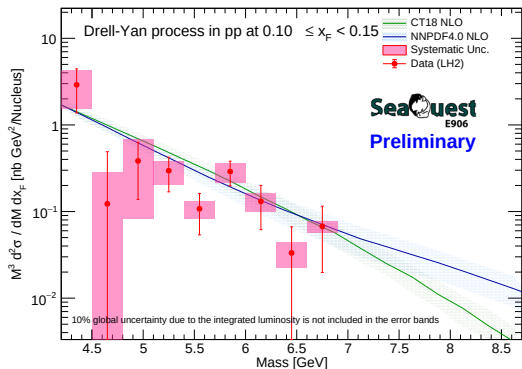


CrossSection LH2 $0.10 \leq x_F < 0.15$ (GeoCenter)

RS67

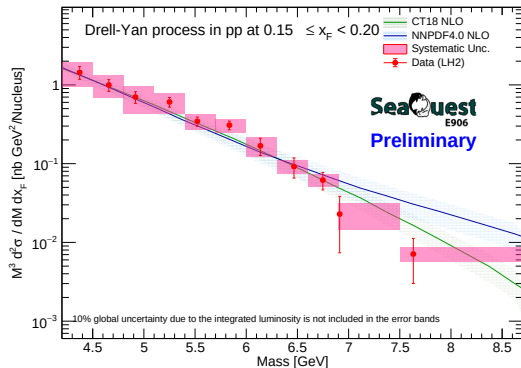


RS70

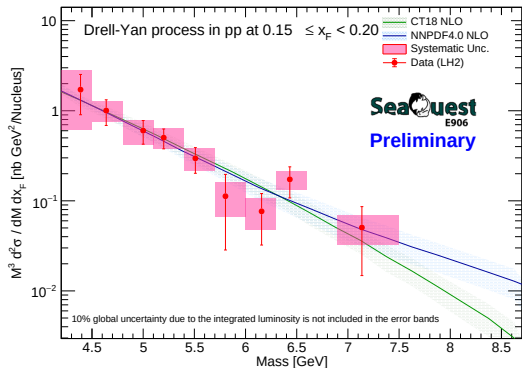


CrossSection LH2 $0.15 \leq x_F < 0.20$ (Centroid)

RS67

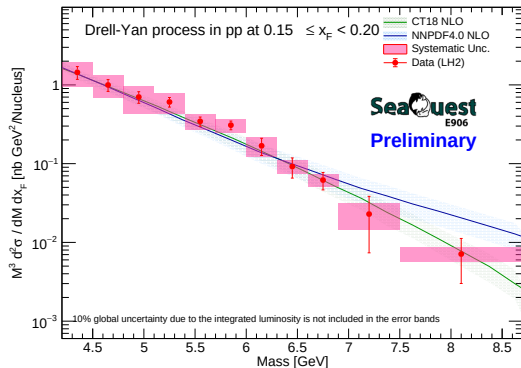


RS70

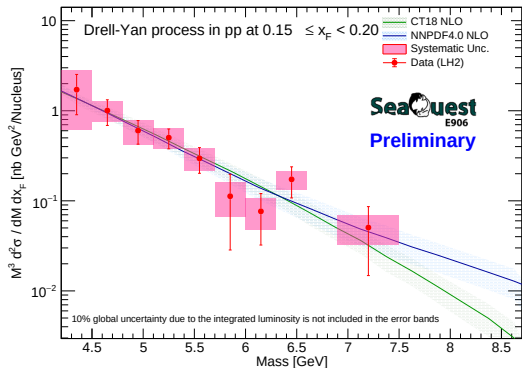


CrossSection LH2 $0.15 \leq x_F < 0.20$ (GeoCenter)

RS67

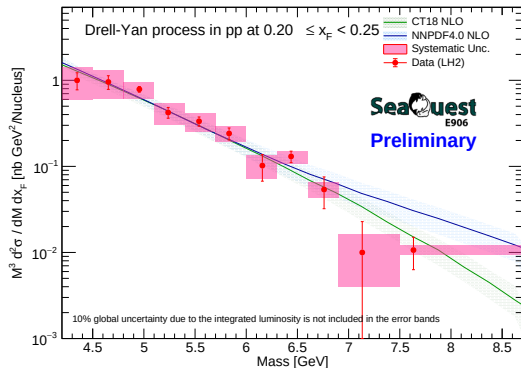


RS70

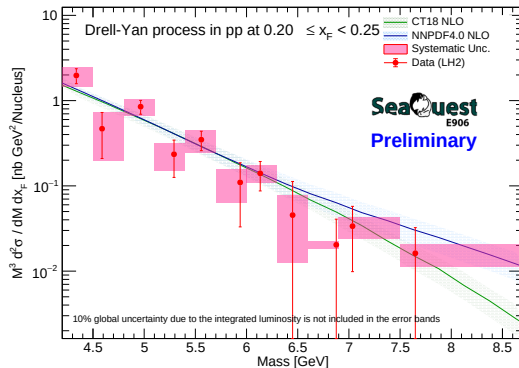


CrossSection LH2 $0.20 \leq x_F < 0.25$ (Centroid)

RS67

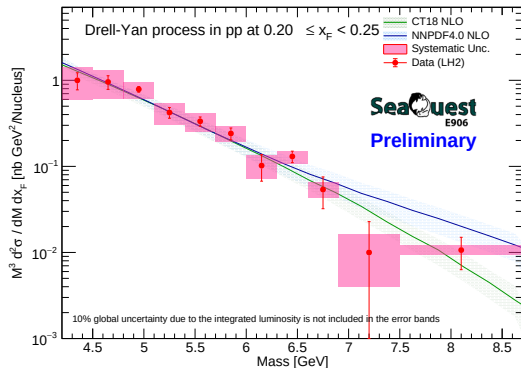


RS70

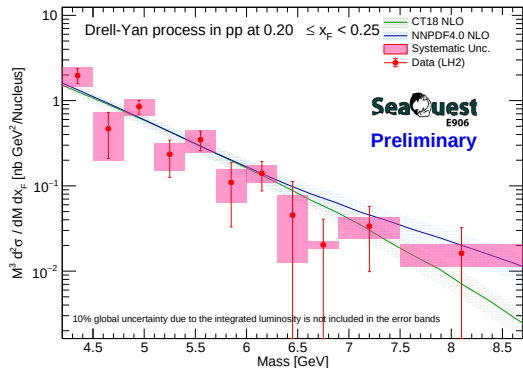


CrossSection LH2 $0.20 \leq x_F < 0.25$ (GeoCenter)

RS67

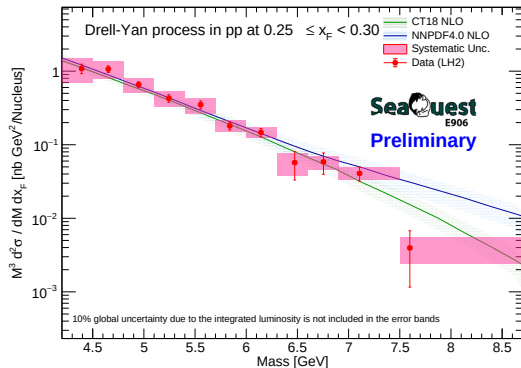


RS70

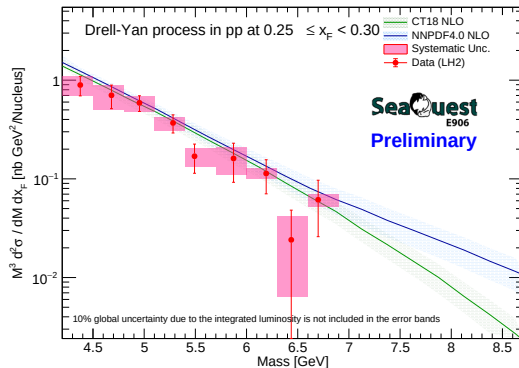


CrossSection LH2 $0.25 \leq x_F < 0.30$ (Centroid)

RS67

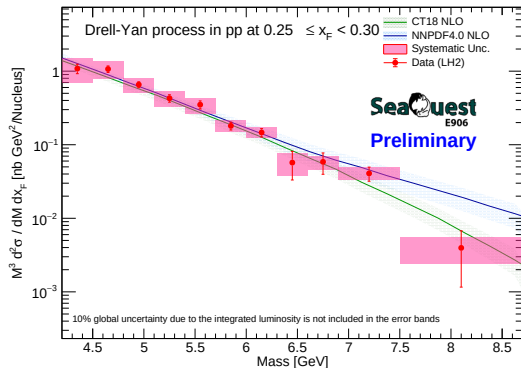


RS70

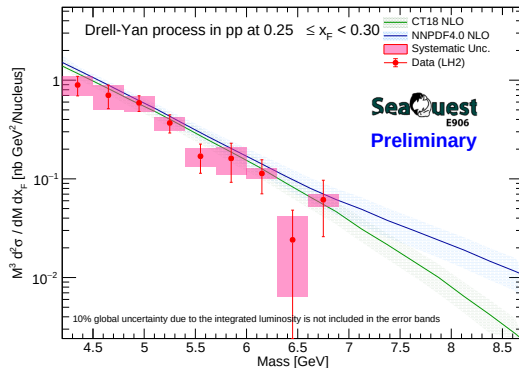


CrossSection LH2 $0.25 \leq x_F < 0.30$ (GeoCenter)

RS67

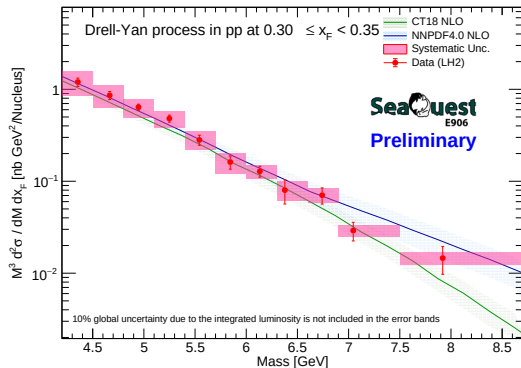


RS70

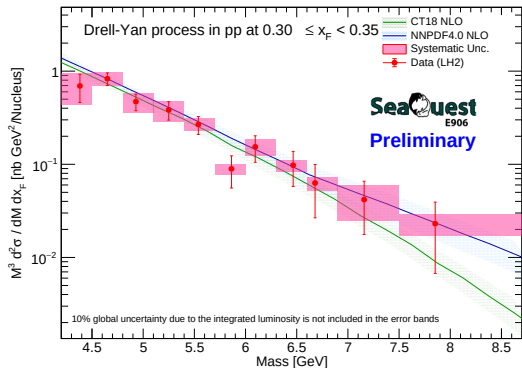


CrossSection LH2 $0.30 \leq x_F < 0.35$ (Centroid)

RS67

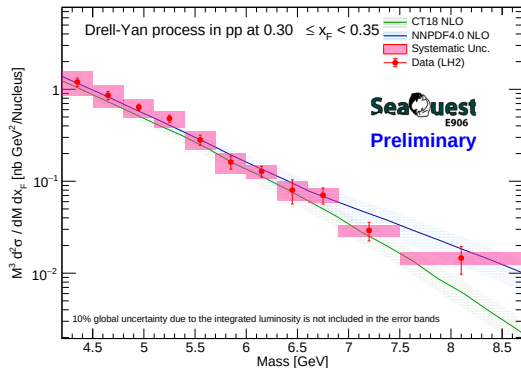


RS70

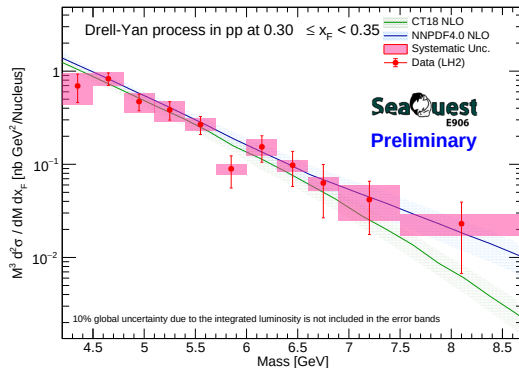


CrossSection LH2 $0.30 \leq x_F < 0.35$ (GeoCenter)

RS67

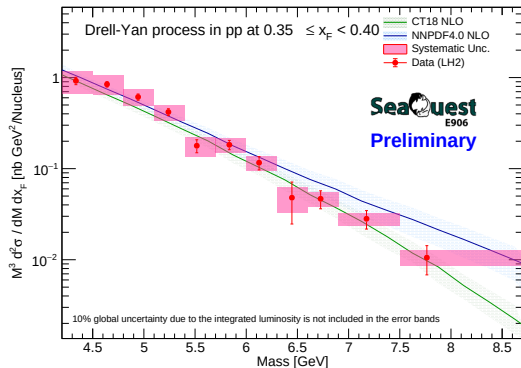


RS70

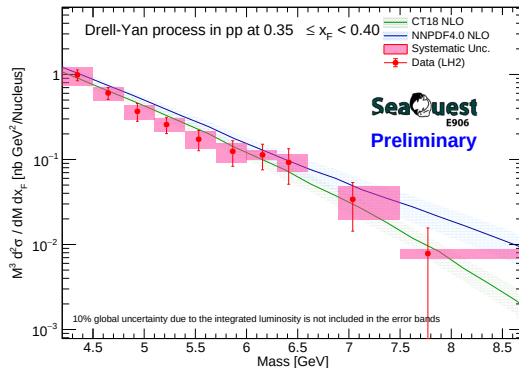


CrossSection LH2 $0.35 \leq x_F < 0.40$ (Centroid)

RS67

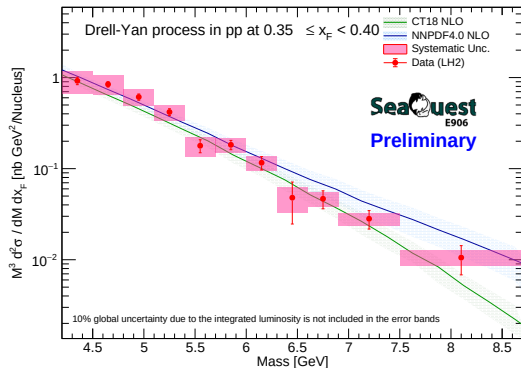


RS70

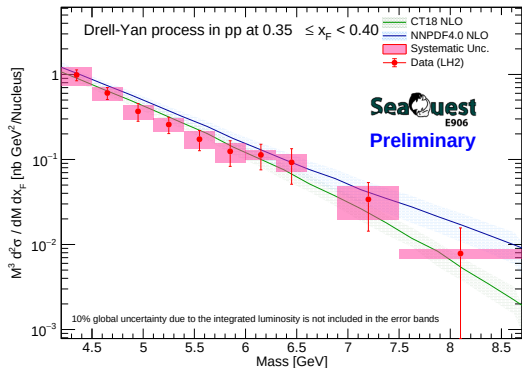


CrossSection LH2 $0.35 \leq x_F < 0.40$ (GeoCenter)

RS67

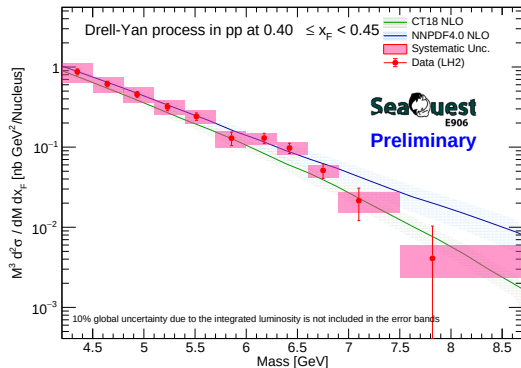


RS70

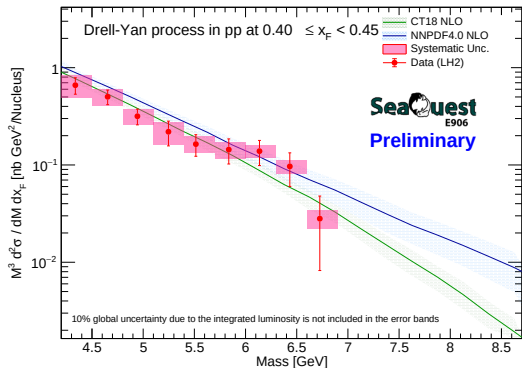


CrossSection LH2 $0.40 \leq x_F < 0.45$ (Centroid)

RS67

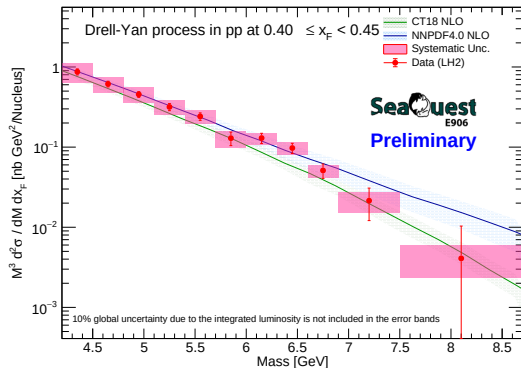


RS70

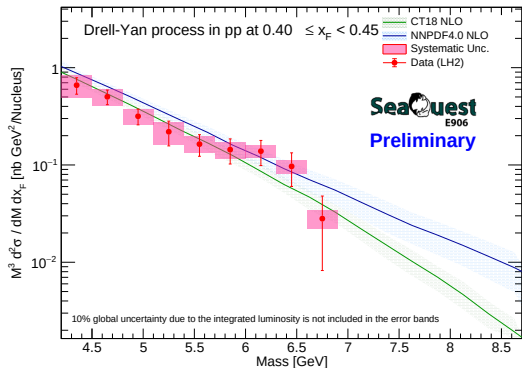


CrossSection LH2 $0.40 \leq x_F < 0.45$ (GeoCenter)

RS67

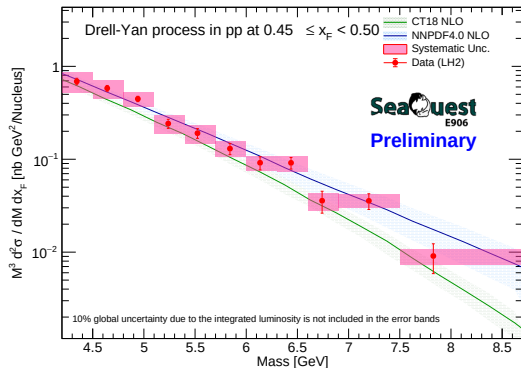


RS70

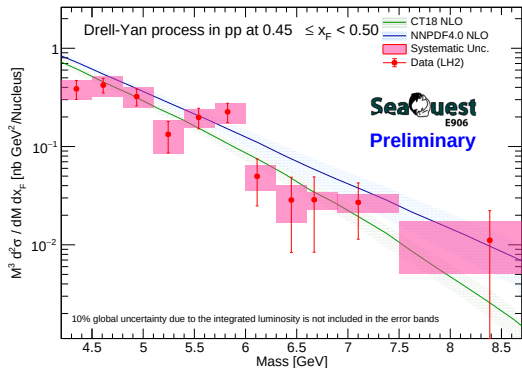


CrossSection LH2 $0.45 \leq x_F < 0.50$ (Centroid)

RS67

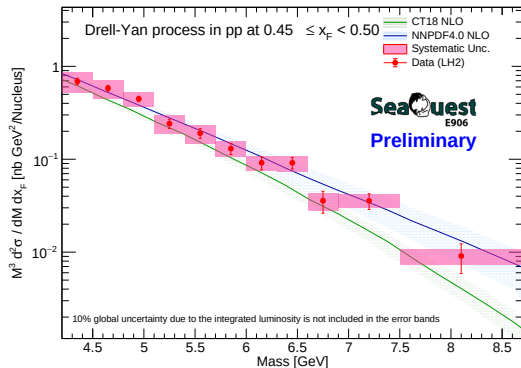


RS70

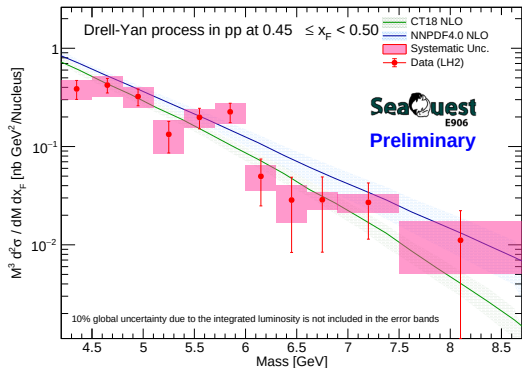


CrossSection LH2 $0.45 \leq x_F < 0.50$ (GeoCenter)

RS67

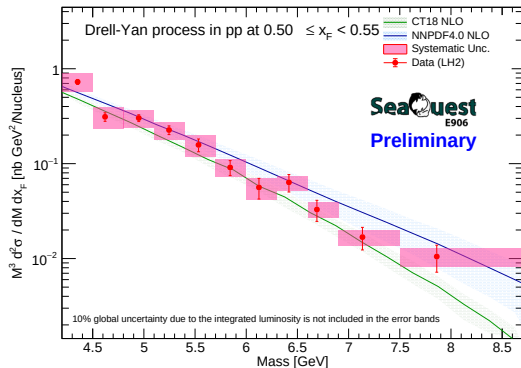


RS70

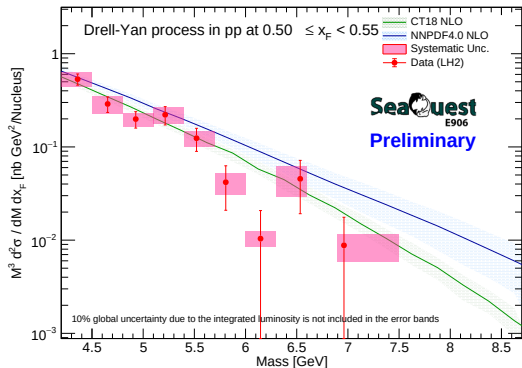


CrossSection LH2 $0.50 \leq x_F < 0.55$ (Centroid)

RS67

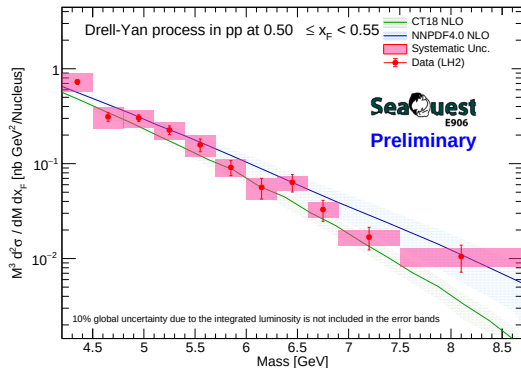


RS70

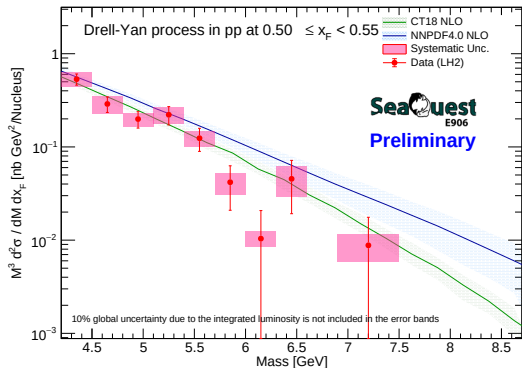


CrossSection LH2 $0.50 \leq x_F < 0.55$ (GeoCenter)

RS67

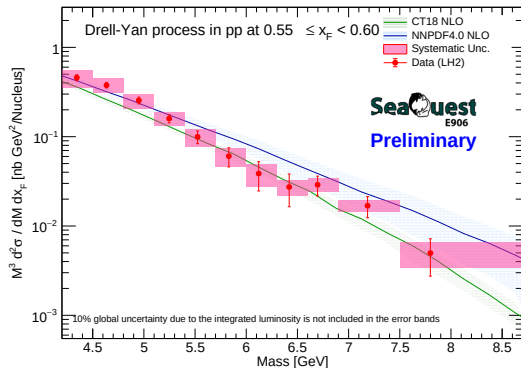


RS70

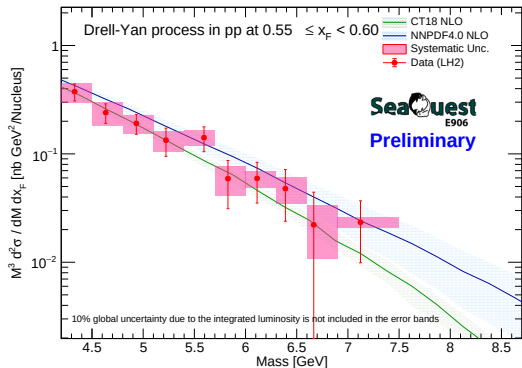


CrossSection LH2 $0.55 \leq x_F < 0.60$ (Centroid)

RS67

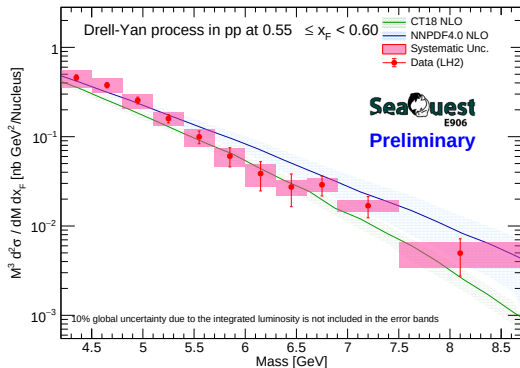


RS70

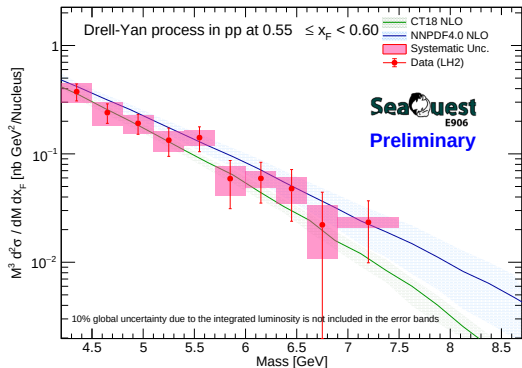


CrossSection LH2 $0.55 \leq x_F < 0.60$ (GeoCenter)

RS67

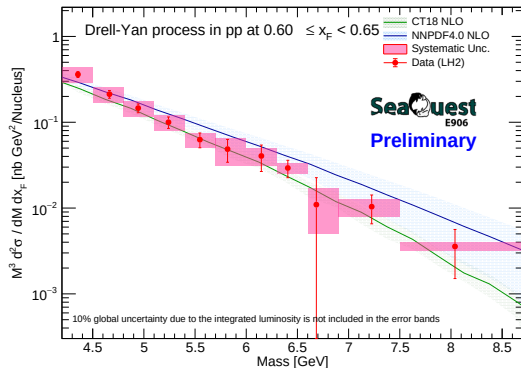


RS70

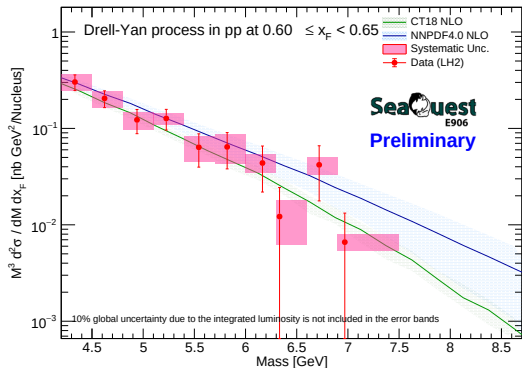


CrossSection LH2 $0.60 \leq x_F < 0.65$ (Centroid)

RS67

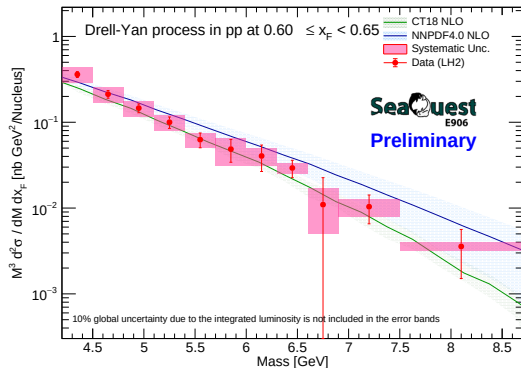


RS70

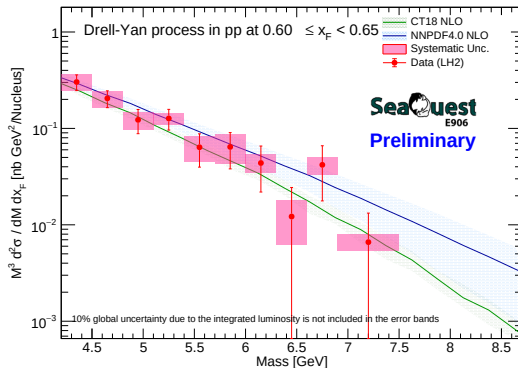


CrossSection LH2 $0.60 \leq x_F < 0.65$ (GeoCenter)

RS67

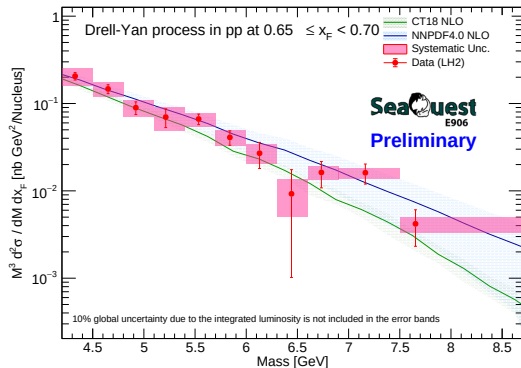


RS70

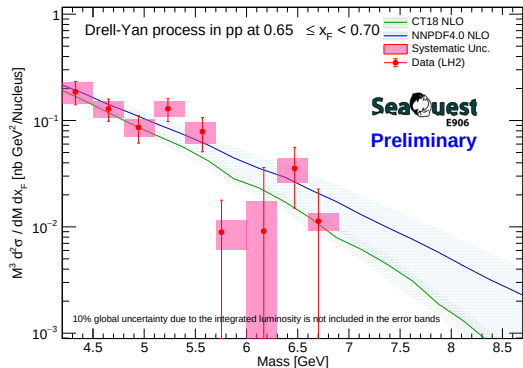


CrossSection LH2 $0.65 \leq x_F < 0.70$ (Centroid)

RS67

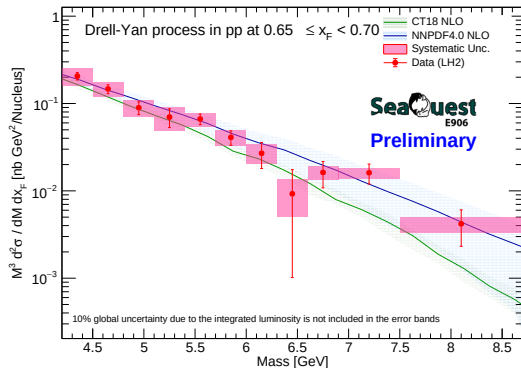


RS70

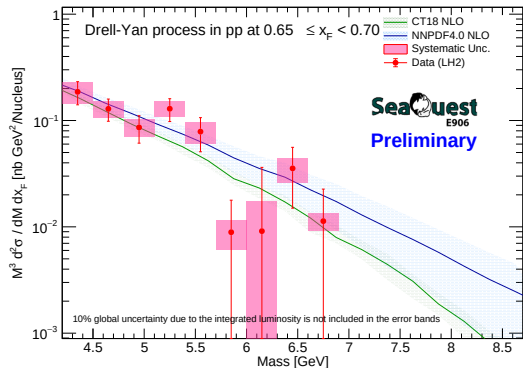


CrossSection LH2 $0.65 \leq x_F < 0.70$ (GeoCenter)

RS67

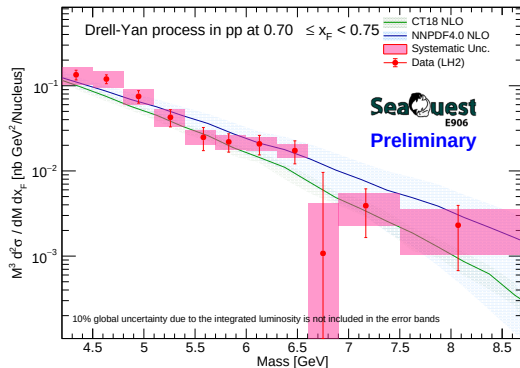


RS70

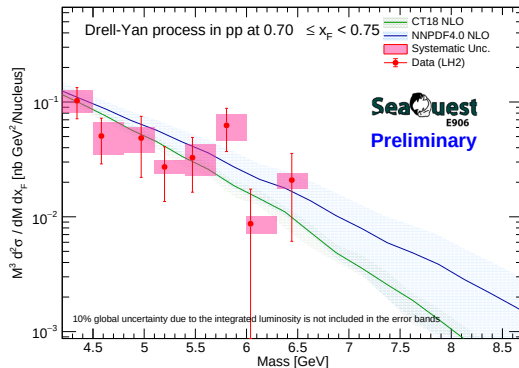


CrossSection LH2 $0.70 \leq x_F < 0.75$ (Centroid)

RS67

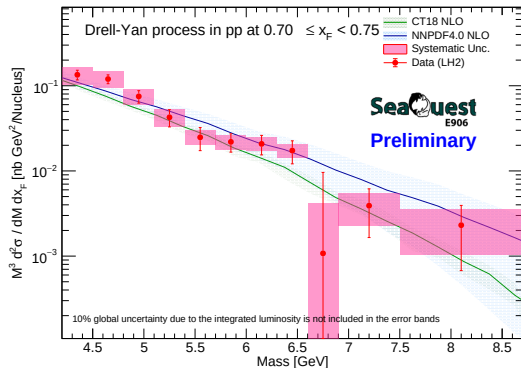


RS70

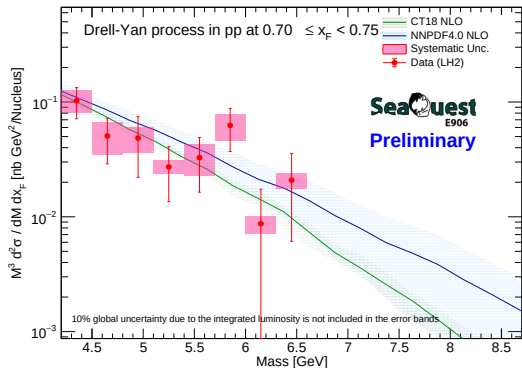


CrossSection LH2 $0.70 \leq x_F < 0.75$ (GeoCenter)

RS67

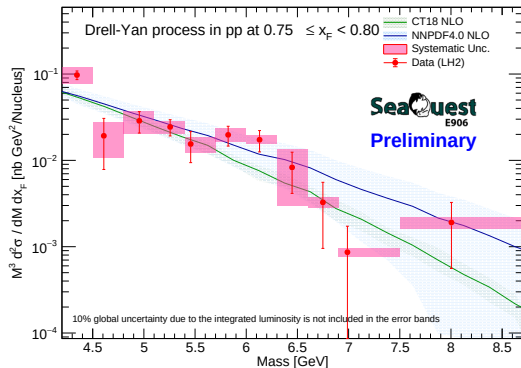


RS70

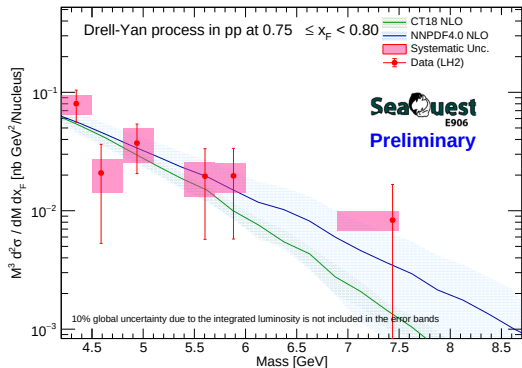


CrossSection LH2 $0.75 \leq x_F < 0.80$ (Centroid)

RS67

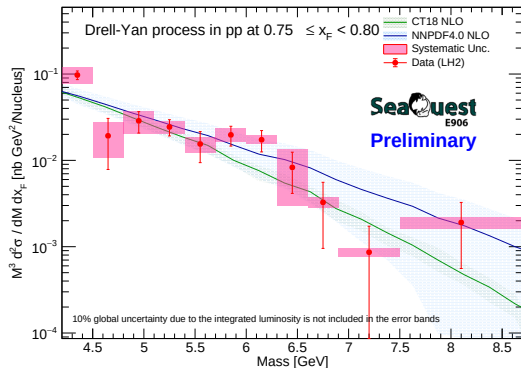


RS70

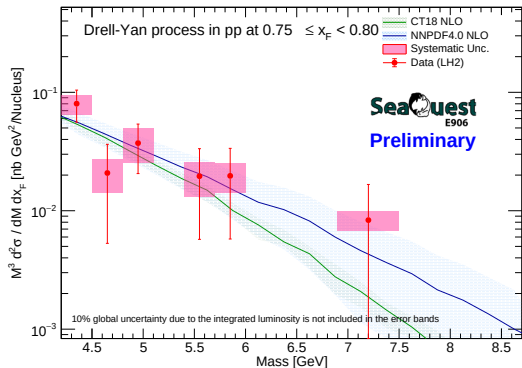


CrossSection LH2 $0.75 \leq x_F < 0.80$ (GeoCenter)

RS67

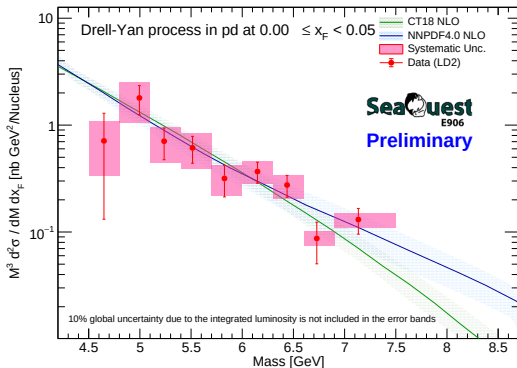


RS70

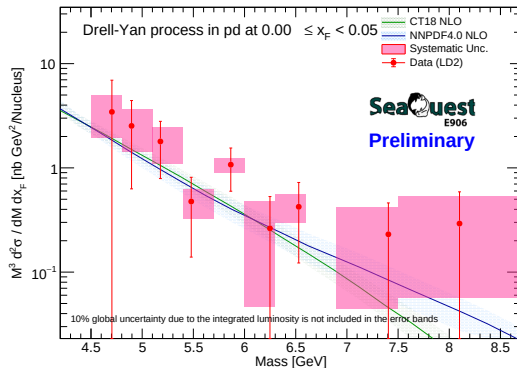


CrossSection LD2 $0.00 \leq x_F < 0.05$ (Centroid)

RS67

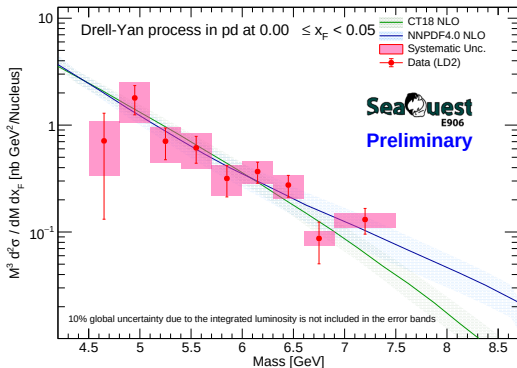


RS70

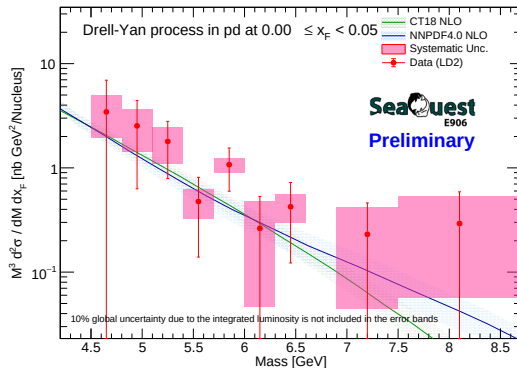


CrossSection LD2 $0.00 \leq x_F < 0.05$ (GeoCenter)

RS67

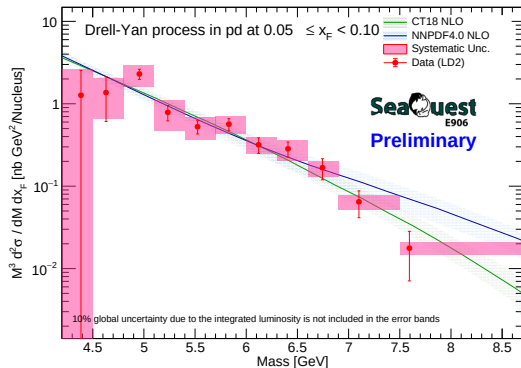


RS70

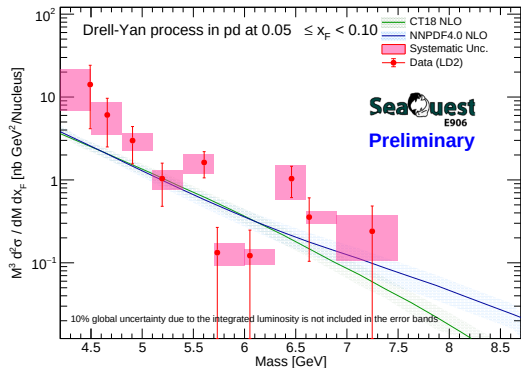


CrossSection LD2 $0.05 \leq x_F < 0.10$ (Centroid)

RS67

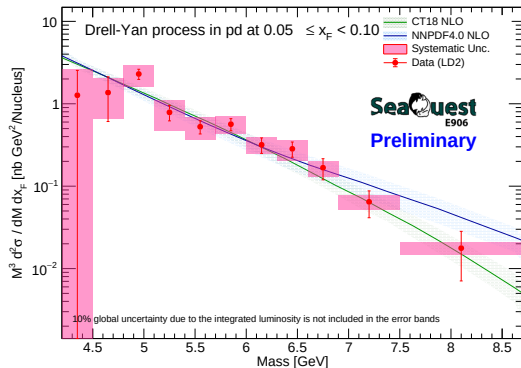


RS70

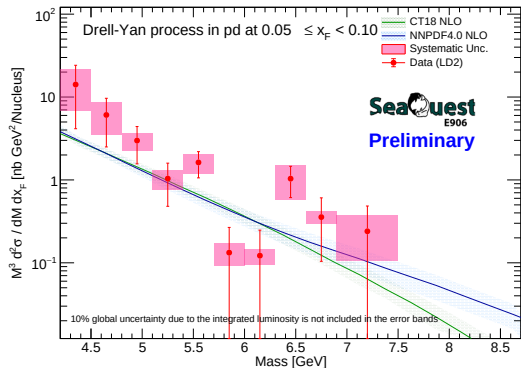


CrossSection LD2 $0.05 \leq x_F < 0.10$ (GeoCenter)

RS67

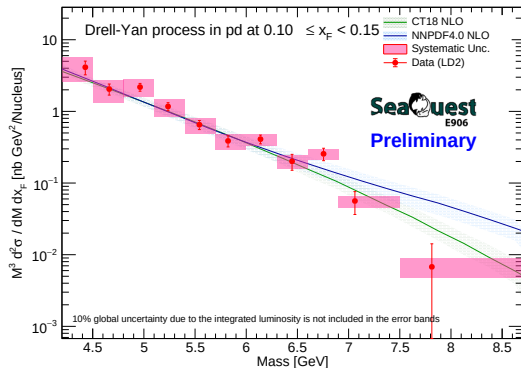


RS70

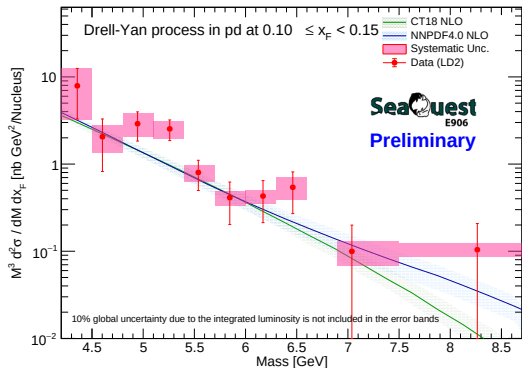


CrossSection LD2 $0.10 \leq x_F < 0.15$ (Centroid)

RS67

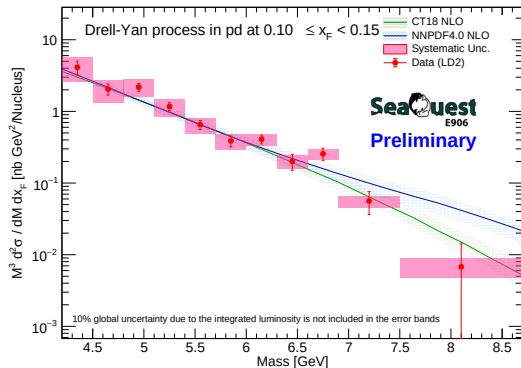


RS70

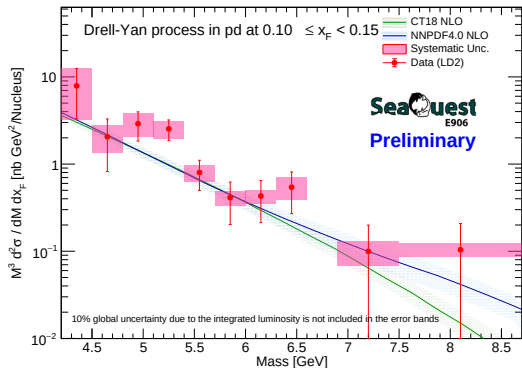


CrossSection LD2 $0.10 \leq x_F < 0.15$ (GeoCenter)

RS67

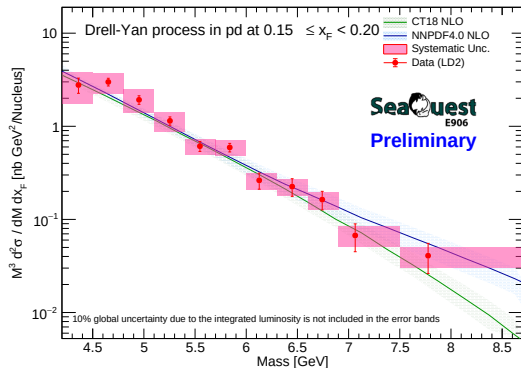


RS70

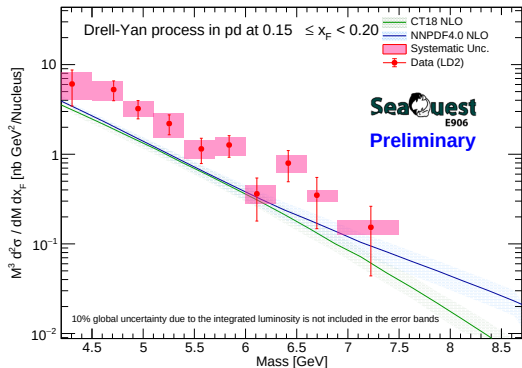


CrossSection LD2 $0.15 \leq x_F < 0.20$ (Centroid)

RS67

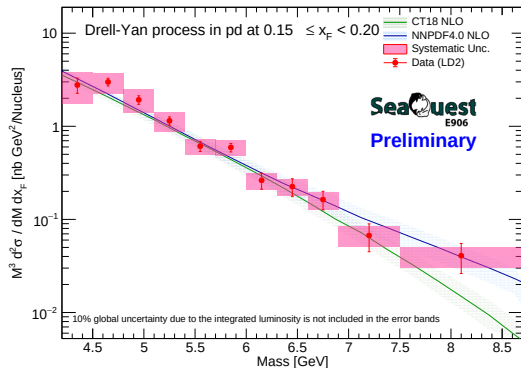


RS70

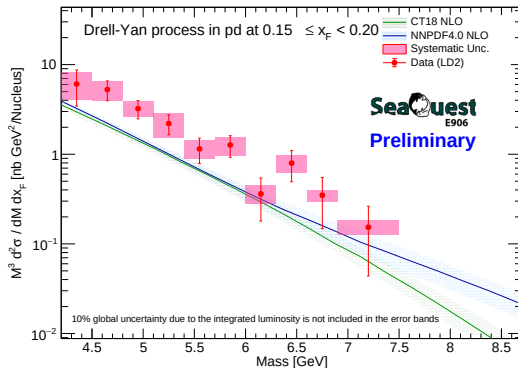


CrossSection LD2 $0.15 \leq x_F < 0.20$ (GeoCenter)

RS67

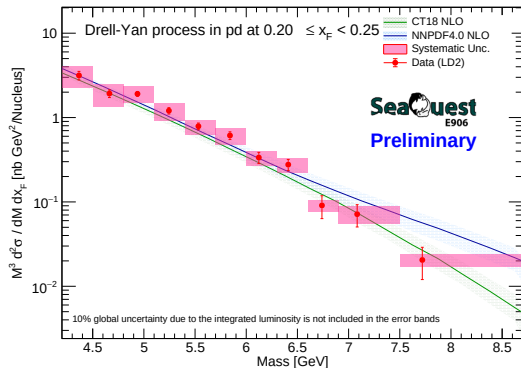


RS70

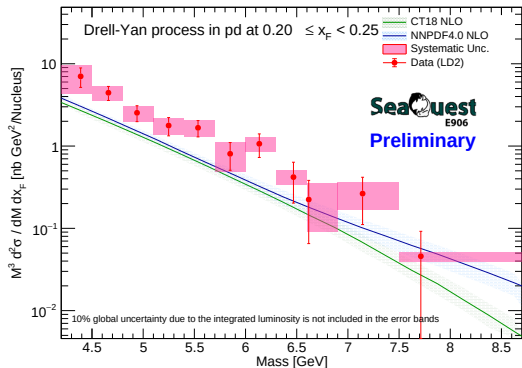


CrossSection LD2 $0.20 \leq x_F < 0.25$ (Centroid)

RS67

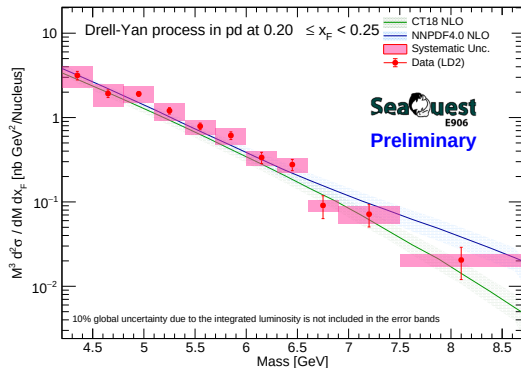


RS70

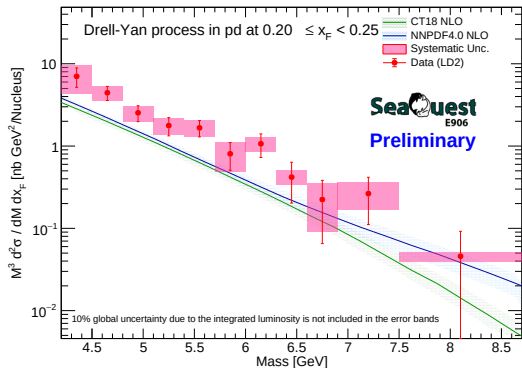


CrossSection LD2 $0.20 \leq x_F < 0.25$ (GeoCenter)

RS67

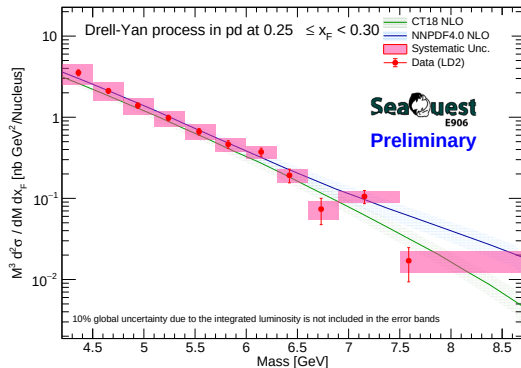


RS70

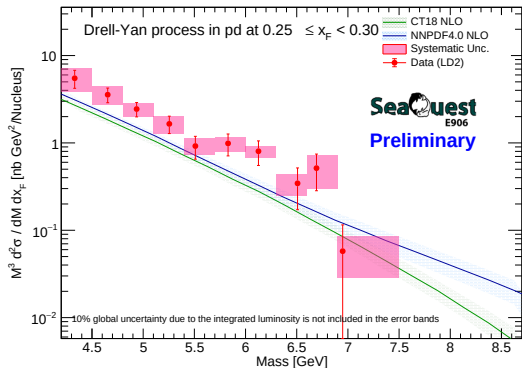


CrossSection LD2 $0.25 \leq x_F < 0.30$ (Centroid)

RS67

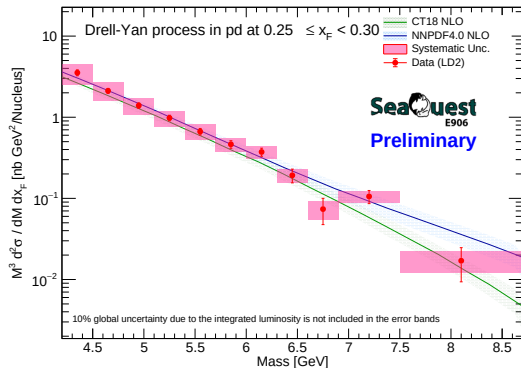


RS70

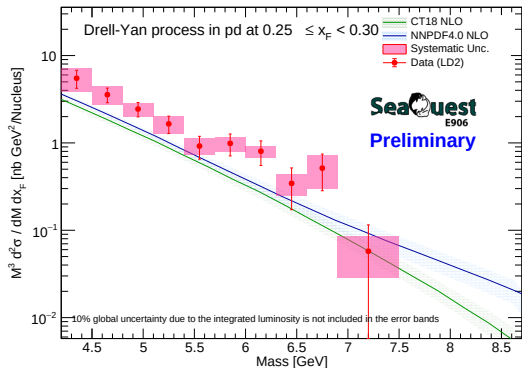


CrossSection LD2 $0.25 \leq x_F < 0.30$ (GeoCenter)

RS67

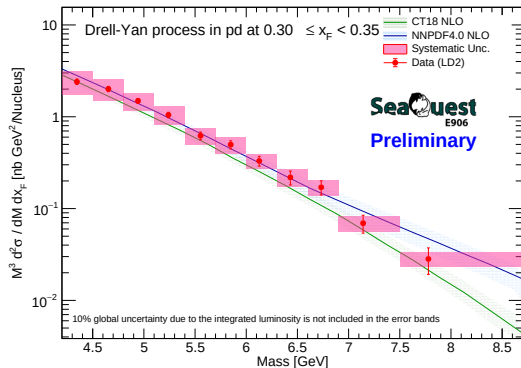


RS70

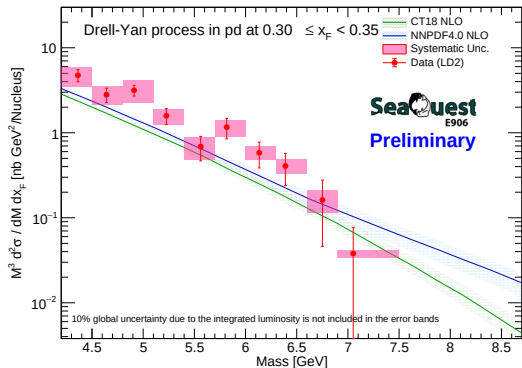


CrossSection LD2 $0.30 \leq x_F < 0.35$ (Centroid)

RS67

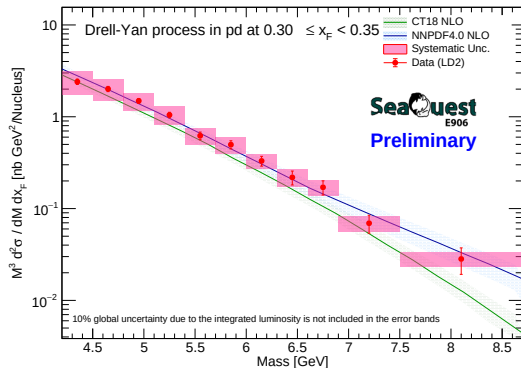


RS70

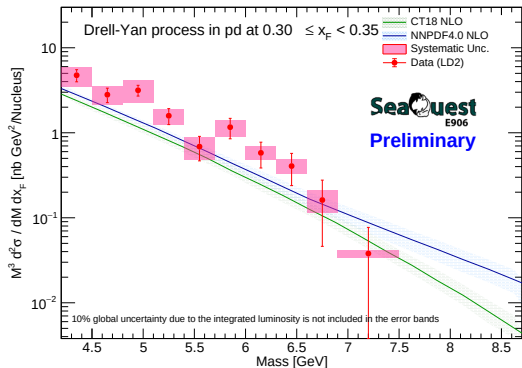


CrossSection LD2 $0.30 \leq x_F < 0.35$ (GeoCenter)

RS67

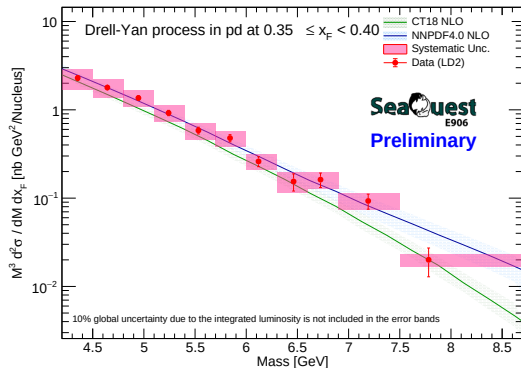


RS70

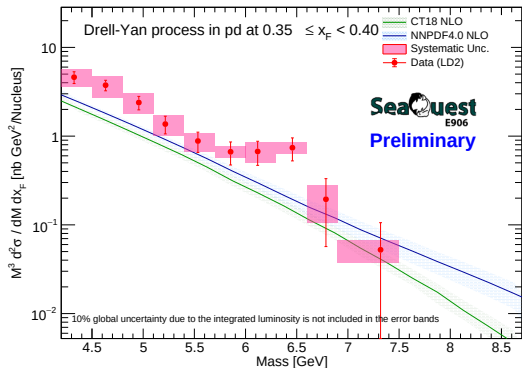


CrossSection LD2 $0.35 \leq x_F < 0.40$ (Centroid)

RS67

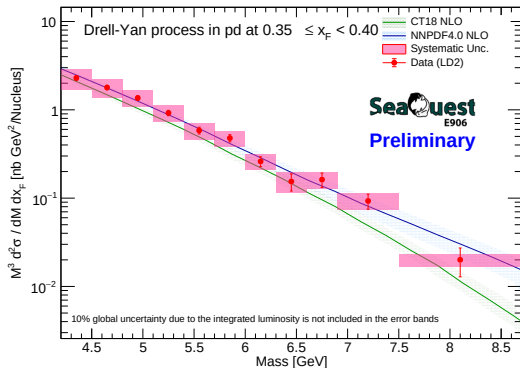


RS70

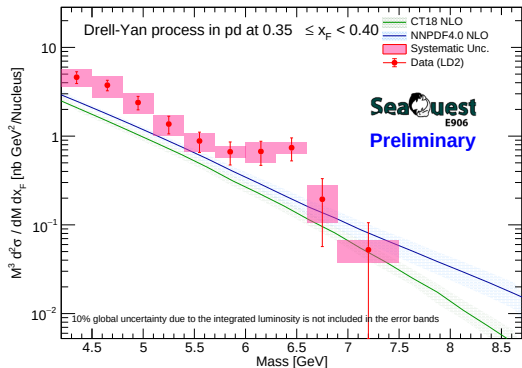


CrossSection LD2 $0.35 \leq x_F < 0.40$ (GeoCenter)

RS67

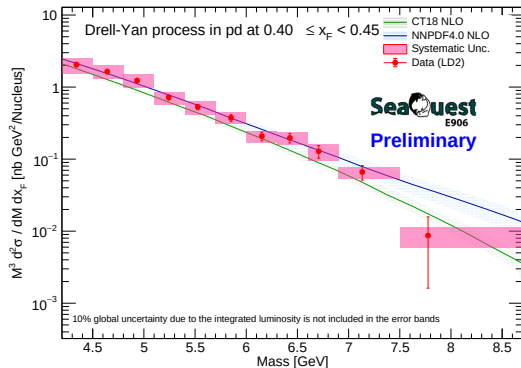


RS70

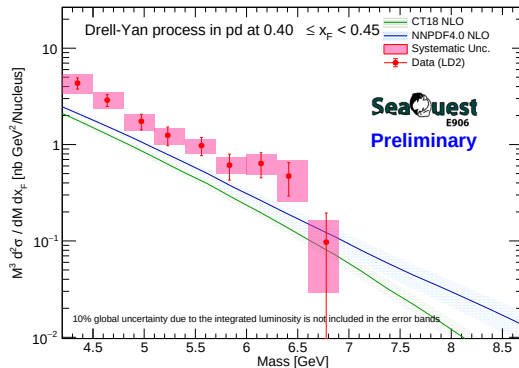


CrossSection LD2 $0.40 \leq x_F < 0.45$ (Centroid)

RS67

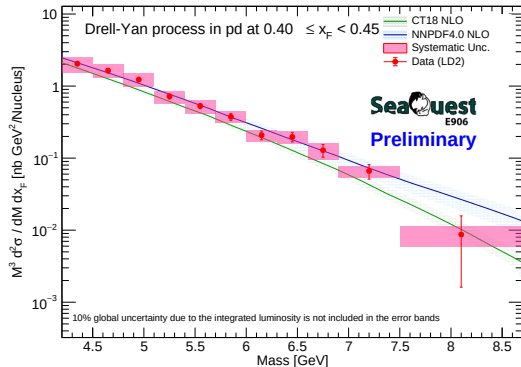


RS70

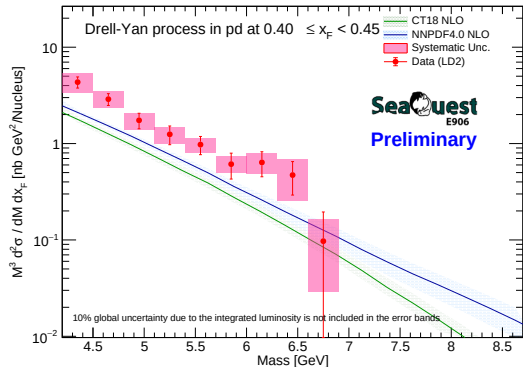


CrossSection LD2 $0.40 \leq x_F < 0.45$ (GeoCenter)

RS67

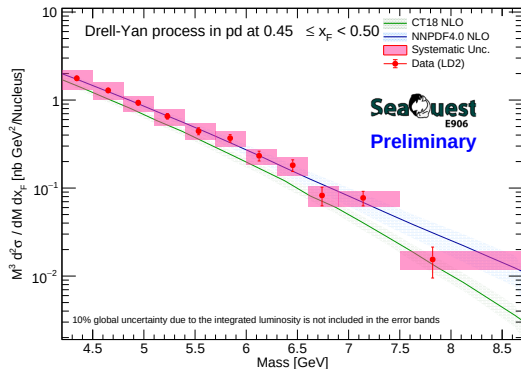


RS70

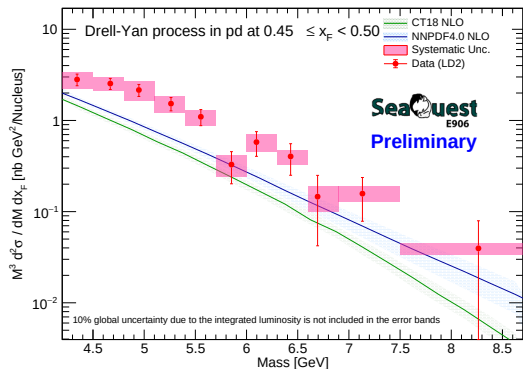


CrossSection LD2 $0.45 \leq x_F < 0.50$ (Centroid)

RS67

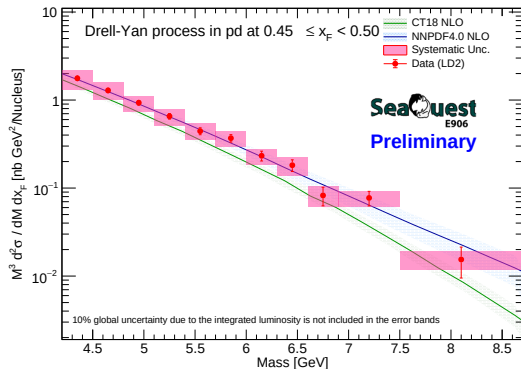


RS70

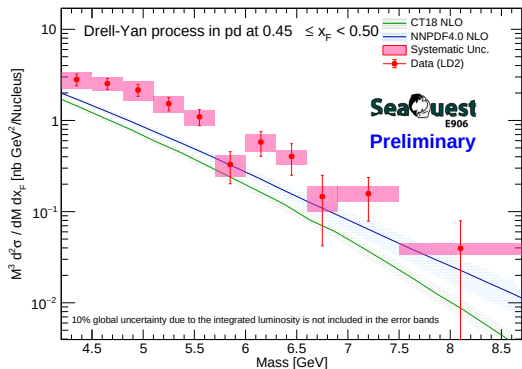


CrossSection LD2 $0.45 \leq x_F < 0.50$ (GeoCenter)

RS67

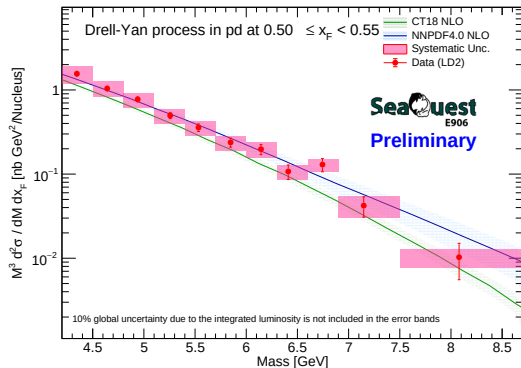


RS70

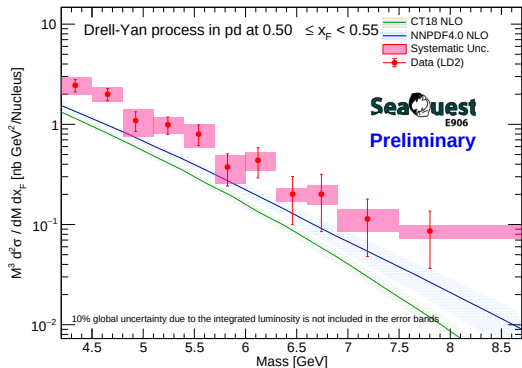


CrossSection LD2 $0.50 \leq x_F < 0.55$ (Centroid)

RS67

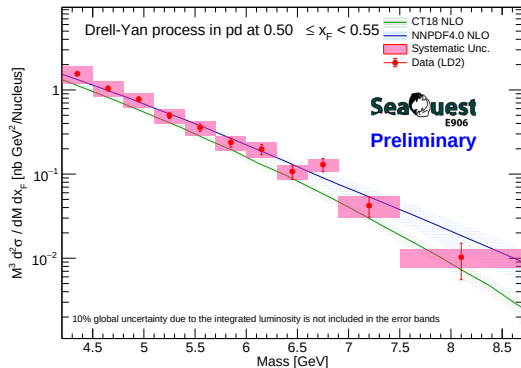


RS70

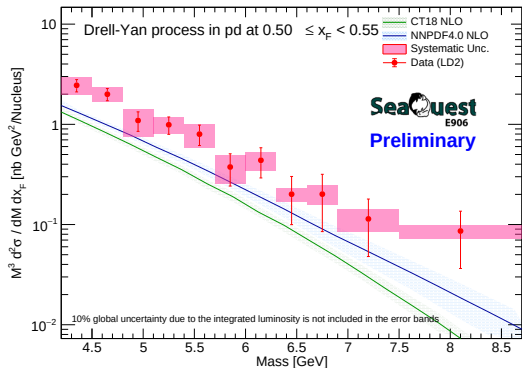


CrossSection LD2 $0.50 \leq x_F < 0.55$ (GeoCenter)

RS67

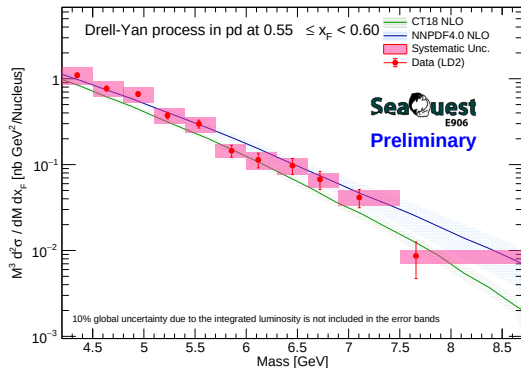


RS70

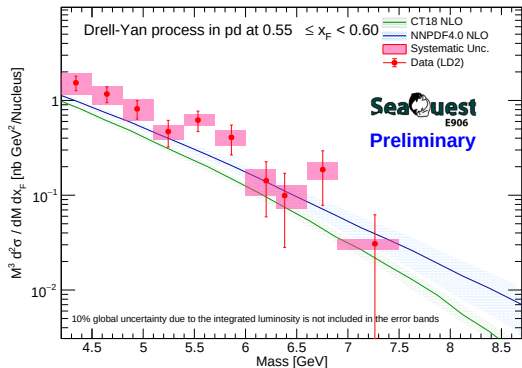


CrossSection LD2 $0.55 \leq x_F < 0.60$ (Centroid)

RS67

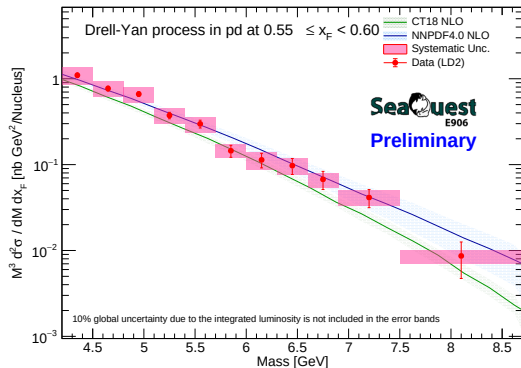


RS70

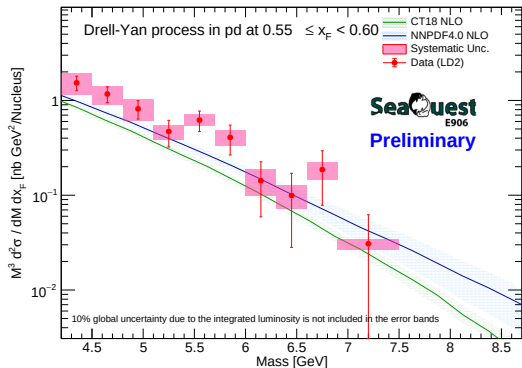


CrossSection LD2 $0.55 \leq x_F < 0.60$ (GeoCenter)

RS67

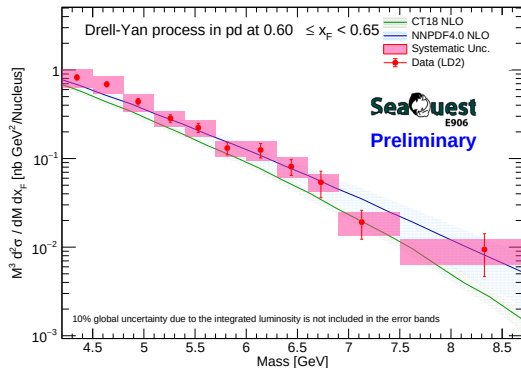


RS70

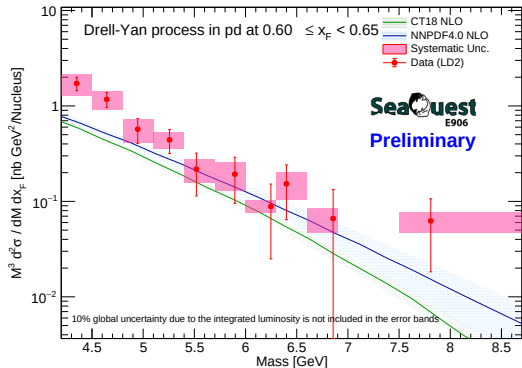


CrossSection LD2 $0.60 \leq x_F < 0.65$ (Centroid)

RS67

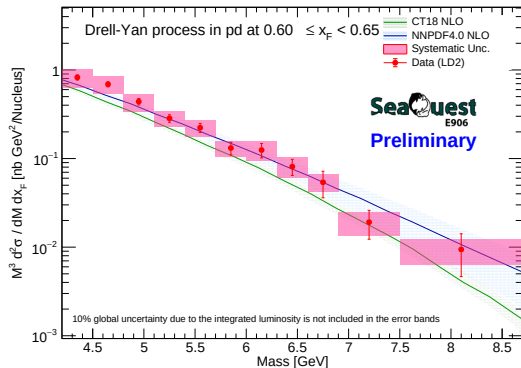


RS70

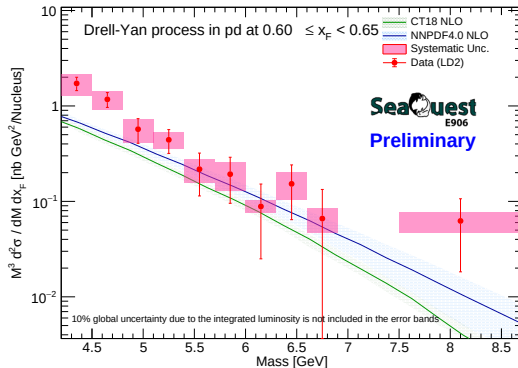


CrossSection LD2 $0.60 \leq x_F < 0.65$ (GeoCenter)

RS67

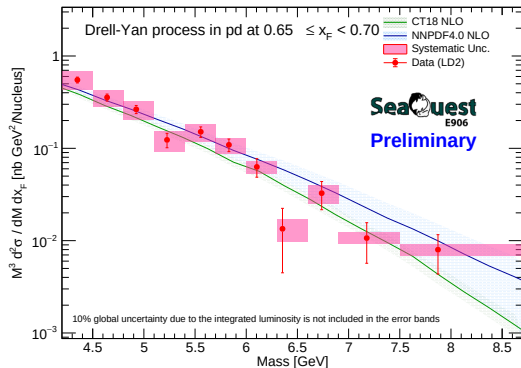


RS70

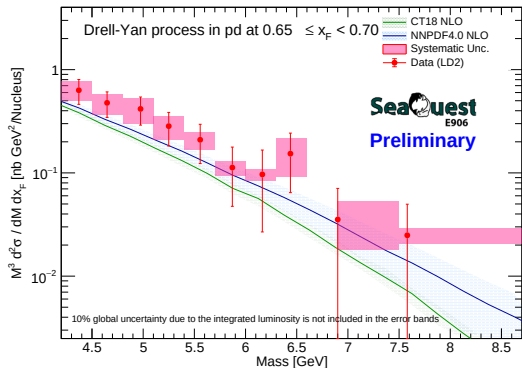


CrossSection LD2 $0.65 \leq x_F < 0.70$ (Centroid)

RS67

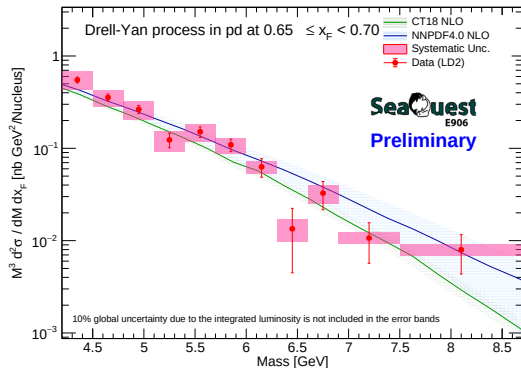


RS70

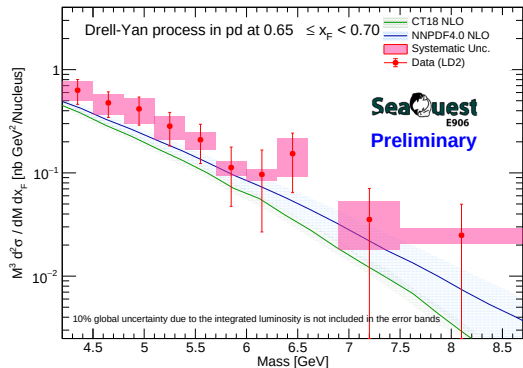


CrossSection LD2 $0.65 \leq x_F < 0.70$ (GeoCenter)

RS67

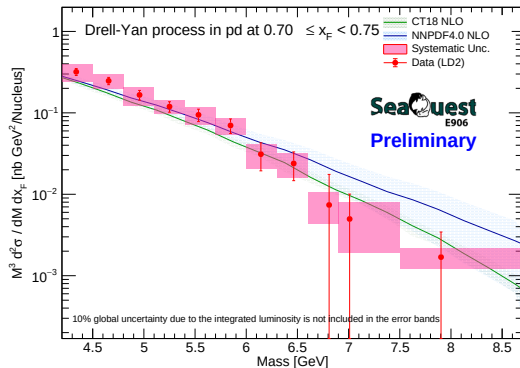


RS70

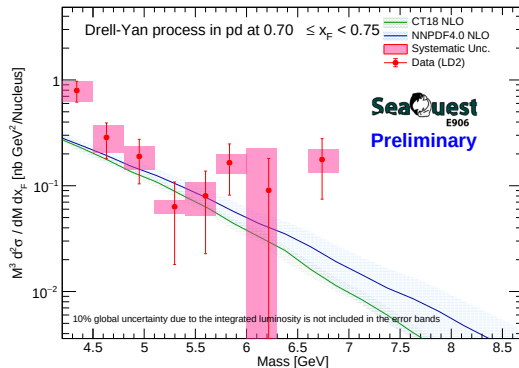


CrossSection LD2 $0.70 \leq x_F < 0.75$ (Centroid)

RS67

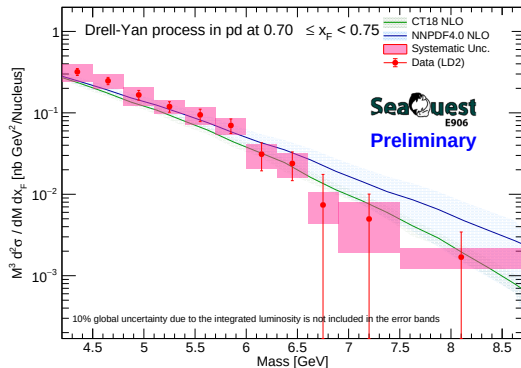


RS70

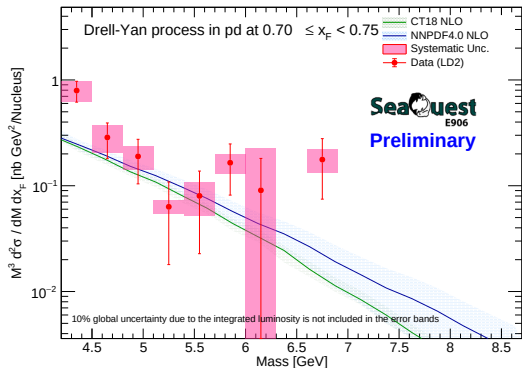


CrossSection LD2 $0.70 \leq x_F < 0.75$ (GeoCenter)

RS67

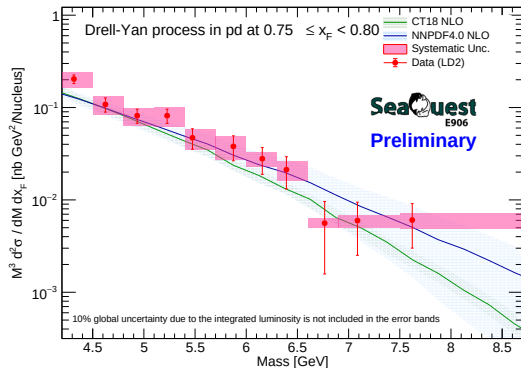


RS70

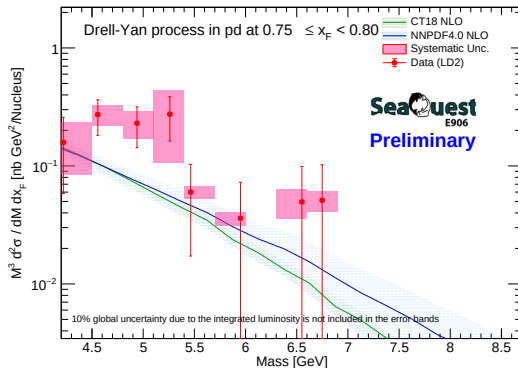


CrossSection LD2 $0.75 \leq x_F < 0.80$ (Centroid)

RS67

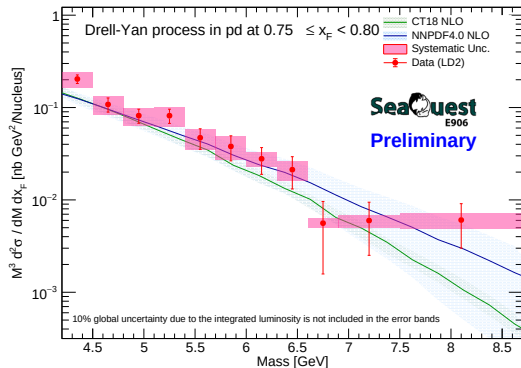


RS70



CrossSection LD2 $0.75 \leq x_F < 0.80$ (GeoCenter)

RS67



RS70

